Transportation Research Part C 191 (2026) 105816

Contents lists available at ScienceDirect

Transportation Research Part C

journal homepage: www.elsevier.com/locate/trc

A data-driven end-to-end framework for traffic estimation with cross-network generalization

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ARTICLE INFO

Keywords:
Traffic estimation
Graph neural networks
Spatio-temporal representation
Transportation network
Contrastive learning
Transportation planning

ABSTRACT

With rapid urbanization, estimating network-wide traffic statistics in urban areas without historical observations has become increasingly important for ensuring efficient transportation systems and improving urban quality of life. Traditionally, trip-based and activity-based travel demand models have been employed to address this problem. While recent advances in machine learning have facilitated network-wide traffic analysis for existing urban regions, data-driven approaches capable of traffic estimation in previously unobserved areas during the planning stage remain limited. To address this gap, we first define two traffic estimation tasks: transductive traffic estimation, where models have partial observations within the same network, and inductive traffic estimation, where models estimate traffic statistics for entirely unseen networks. We then propose Generalizable Traffic Estimation with Graph Contrastive Objective (GTE-GC), a data-driven framework that estimates network-wide traffic speeds using only transportation network topology and land-use characteristics. The model integrates a spatial feed-forward neural network, temporal self-attention, and graph contrastive learning to support both transductive and inductive estimation. Extensive experiments across ten European cities demonstrate that GTE-GC achieves competitive transductive performance and superior inductive generalization, with source-domain sensitivity quantified through empirical cross-city evaluation. Additional data ablation analyses highlight the role of each data source. The proposed framework offers a scalable and efficient tool to complement traditional simulation-based approaches, providing practical support for early-stage transportation planning in emerging urban contexts. The code of this work is publicly available at: https://github.com/yuyolshin-2/GTEGC.

1. Introduction

Urban transportation systems are central to the functionality and livability of modern cities. With the urban population projected to increase by 2.5 billion by 2050 and new cities being built worldwide, there is a growing need for accurate and scalable traffic analysis tools that can be applied to previously unobserved urban environments, such as new cities and regions lacking traffic sensor coverage. Mobility efficiency is recognized as a key determinant of urban quality of life and featured in major city performance indicators, such as the Smart City Index (IMD, 2024), the Global Livability Index (EIU, 2024), and the City Prosperity Index (UN-Habitat, 2016).

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https://doi.org/10.1016/j.trc.2026.105816
Received 14 November 2025; Received in revised form 10 June 2026; Accepted 15 June 2026
Available online 22 June 2026
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A major challenge in this context is the network-wide estimation of traffic conditions in newly developed or data-scarce transportation networks where no historical traffic observations are available. Accurate traffic estimation can support applications, including infrastructure planning (Narayanaswami, 2017), travel time estimation (Zhang et al., 2019b), and accident hotspot identification (Sajed et al., 2019). Although recent advances in data-driven machine learning have yielded promising results for analyzing dynamic traffic data in operational contexts (Yao et al., 2018; Bai et al., 2019; Wang et al., 2023; Xu et al., 2024; Shin and Yoon, 2024; Zhang et al., 2025), their application in early-stage urban planning for regions with no prior data remains limited.

Traditional travel demand models have long served as the foundational framework for estimating network-wide traffic demand, speed, flow, and travel time (NCHRP, 2012). These models are generally divided into two categories: trip-based four-step models (FSMs) and activity-based models (ABMs). FSMs operate at the zonal level, generating aggregated travel demand through a sequential process comprising trip generation, trip distribution, mode choice, and traffic assignment. Each stage integrates diverse data sources, such as land use, built environment, and demographic characteristics, to simulate travel behavior and traffic patterns (Calabrese et al., 2013; Tillema et al., 2006; Lo et al., 2004; Bliemer et al., 2017). In contrast, ABMs were developed to address the limitations of FSMs by modeling travel behavior at the individual tour level. By defining a tour as a sequence of trips that begins and ends at the same anchor location and reflects activity chaining, ABMs capture interdependencies between trips within a tour, allowing for more behaviorally realistic simulations (Castiglione et al., 2015). Despite their contributions, however, both FSMs and ABMs face notable challenges. Their multi-stage structures are prone to error propagation, where inaccuracies in early stages can adversely affect downstream outputs. Moreover, both approaches often lack access to reliable real-world data necessary for training, validation, and calibration of models. In addition, ABMs can be computationally intensive, making them challenging to scale across diverse geographic contexts.

On the other hand, machine learning (ML) and artificial intelligence (AI) techniques have emerged as solutions in end-to-end, network-wide traffic analysis tasks, demonstrating strong performance in traffic estimation and prediction (Li et al., 2018; Yao et al., 2018; Liu et al., 2022; Thodi et al., 2022; Abdelraouf et al., 2023; Shin and Yoon, 2024; Zhou et al., 2025a,b), missing data imputation (Kong et al., 2023; Wei et al., 2024), and traffic assignment (Hu and Xie, 2025; Liu and Meidani, 2024) by learning spatio-temporal patterns from dynamic, multi-source traffic datasets. However, there have been limited efforts to address the challenge of estimating network-wide traffic statistics in urban regions without prior knowledge of historical data. Most studies aim to predict short-term traffic based on the available historical observations and tend to overfit to local spatial patterns, limiting generalizability to new or unobserved urban regions. This restricts their utility in planning contexts, where traffic statistics must often be estimated in environments with no prior traffic observations or sensor infrastructure.

To address these challenges, we formulate and investigate two traffic estimation tasks with end-to-end data-driven models in this study: *transductive* and *inductive traffic estimation*, which require generalization in partially observed and entirely unobserved networks, respectively.

- *Transductive Traffic Estimation* refers to estimating traffic conditions on road segments within the same transportation network used for training. In this setting, the network of a single city is partitioned into mutually exclusive training and test subsets, allowing the model to learn from a portion of the city while being evaluated on the remaining, unobserved segments. In this task, we can assume that the training and testing sets are samples from the same underlying distribution, as they are drawn from the same transportation network.
- *Inductive Traffic Estimation* involves predicting traffic conditions in an entirely different transportation network from the one used during training. The model is trained on the full network of one city and evaluated on the complete network of another city, which may differ significantly in urban form, road topology, and land-use characteristics.

Since structural patterns and land-use characteristics vary across different cities, the inductive task poses a greater challenge, as it requires a model to generalize to unfamiliar urban forms and road topologies without any exposure to the target city's data during the training phase. To explicitly clarify the scope and resolution of these tasks, our framework estimates recurring traffic patterns (i.e., Annual Average Daily Traffic, AADT) at the individual road segment level. Specifically, our framework is trained to estimate a characteristic 18-h (5am-11pm) speed profile at an hourly resolution for an average weekday for each individual road segment in the network. We present detailed explanations in Sections 4 and 5.

In this study, we propose GTE-GC (**G**eneralizable **T**raffic **E**stimation with **G**raph **C**ontrastive objective) to address both traffic estimation tasks, which leverages (1) a multi-source dataset comprising traffic speed, transportation network, and land-use data, and (2) a graph-based deep learning architecture tailored to spatio-temporal representation learning. The model captures the spatio-temporal relationships between observed traffic statistics and the structural properties of the transportation network, including segment connectivity, length, and road class. To equip the model's lightweight spatial encoder with sufficient structural context, we explicitly incorporate engineered spatial features-such as the number of nearby segments-into the input data. These features provide complementary cues about the transportation network's density and composition in Euclidean space that cannot be readily inferred from graph connectivity alone. Furthermore, incorporating land-use data enhances the model's ability to interpret the functional roles of road segments, particularly in unobserved networks lacking historical traffic data.

Crucially, while both the transductive and inductive tasks aim to map these static features to traffic speed profiles, they both face challenges related to distribution shifts. Standard message-passing Graph Neural Networks (GNNs) explicitly propagate features across the graph, which inherently encourages overfitting to the specific topological patterns of the training network. This strict structural reliance increases sensitivity to distribution shifts, severely limiting out-of-distribution (OOD) generalization. To overcome this, GTE-GC bypasses explicit message passing during inference. Instead, the neural network is trained using a joint objective that combines a supervised regression loss with a graph-based contrastive loss (Hu et al., 2021), which encourages semantically similar nodes to

produce representations that are closer in the latent space while separating dissimilar ones. This learning strategy uniquely supports both estimation tasks: for the transductive task, it injects essential topological awareness during training without the computational overhead of strict message passing; for the inductive task, the predictor remains lightweight and feature-driven at inference time, preventing overfitting to the source city's graph structure. Ultimately, by coupling complementary spatial features with a contrastive learning objective, GTE-GC achieves highly robust representations that generalize effectively across diverse urban networks.

The main contributions of our study are threefold:

- We formulate two traffic estimation problems: *transductive traffic estimation*, where the model estimates traffic on unobserved subnetworks within the same geospatial context used for training, and *inductive traffic estimation*, where the model generalizes to entirely unseen networks with different geospatial contexts.
- We devise a novel neural network framework, GTE-GC (Generalizable Traffic Estimation with Graph Contrastive objective), that consists of an input encoder, spatio-temporal layers, and an output layer. We also introduce a graph contrastive learning objective that promotes semantic similarity between spatially connected nodes. The contrastive objective enables the model to learn generalizable feature representations while achieving high computational efficiency through the removal of explicit message-passing operation.
- We conduct extensive experiments in ten European cities to validate the effectiveness of GTE-GC. Baseline comparisons reveal that the GTE-GC framework yields competitive results in the transductive task and outperforms baselines in the inductive setting. A data ablation study further highlights the importance of incorporating both network structure and land-use characteristics.

The remainder of the paper is organized as follows. Section 2 reviews related work on data-driven traffic analysis and traditional travel modeling approaches. Section 3 defines the problem and introduces the data, while Section 4 presents the proposed methodology. Experimental settings and results are discussed in Section 5, followed by a detailed discussion in Section 6. Finally, Section 7 concludes the paper and outlines potential for future studies.

2. Literature review

In this section, we highlight existing research efforts, identify current gaps, and explain how the traffic estimation tasks in this study address these gaps. Section 2.1 focuses on data-driven end-to-end machine learning models in network-wide traffic analysis. Section 2.2 reviews model-driven travel demand models and explains how machine learning approaches can be utilized for end-to-end traffic estimation tasks.

2.1. Data-driven end-to-end machine learning approaches for traffic analysis

With the proliferation of sensor networks, GPS traces, and large-scale mobility datasets, research has increasingly focused on data-driven methods for directly estimating and forecasting traffic statistics at the road segment level based on historical observations. Over the past decade, deep learning approaches have gained particular attention due to their ability to process large volumes of data and model complex, nonlinear relationships (Wang et al., 2019; Veres and Moussa, 2020; Shin and Yoon, 2023). Unlike traditional travel demand models, which rely on stepwise demand modeling, these methods infer traffic speed, flow, or density directly from empirical observations, bypassing intermediate modeling steps.

To model the temporal dynamics of traffic data, initial efforts primarily relied on Recurrent Neural Networks (RNNs). Ma et al. (2015) applied a Long Short-Term Memory (LSTM) network to forecast traffic speed and flow, demonstrating superior performance compared to traditional statistical and machine learning models. Subsequent research has typically modeled temporal correlations using RNNs (Li et al., 2018; Zhang et al., 2019a; Yao et al., 2018; Liu et al., 2022; Kong et al., 2023; Abdelraouf et al., 2023; Hu et al., 2025). However, traditional RNNs inherently suffer from limitations such as vanishing gradients and error accumulation. To overcome these challenges, Liu et al. (2022) proposed a novel RNN approach that explicitly incorporates hidden states from multiple historical time steps into the computation of each current time unit. Alternatively, researchers have utilized 1D temporal convolutions to model temporal correlations in traffic data (Yu et al., 2018). To expand the receptive field for the temporal convolution, dilated causal convolution layers are frequently employed (Wu et al., 2019; Yuan et al., 2023; Shin and Yoon, 2024). More recently, transformer architectures have been widely adopted in spatio-temporal models to capture long-range temporal dependencies (Zheng et al., 2020; Guo et al., 2022; Zhang et al., 2022; Huang et al., 2022; Li et al., 2024; Xu et al., 2024; Wei et al., 2024; Hu et al., 2025). Guo et al. (2022) and Li et al. (2024) implemented trend-aware self-attention, which employs 1D convolutions during query, key, and value generation to capture local time-series trends. Furthermore, Xu et al. (2024) proposed a prompter mechanism that enables the model to adaptively adjust its parameters based on evolving temporal patterns.

For spatial modeling, convolutional neural networks (CNNs) were initially explored to capture spatial correlations among adjacent regions (Yao et al., 2018; Bai et al., 2019; Ma et al., 2017; Rempe et al., 2022). However, because CNNs are inherently limited to Euclidean grids, the focus shifted toward graph-based representations of urban transportation networks (Li et al., 2018; Zheng et al., 2020; Shin and Yoon, 2024; Yao et al., 2018; Wu et al., 2021; Liu et al., 2022; Nie et al., 2023; Wei et al., 2024; Kong et al., 2023; Wang et al., 2023; Xu et al., 2024; Zhang et al., 2025; Abdelraouf et al., 2023; Hu et al., 2025; Xu et al., 2020; Zhou et al., 2025a,b). For example, Li et al. (2018) modeled vehicle flow as a diffusion process on graphs using a random-walk-based transition matrix, while recent works (Guo et al., 2022; Kong et al., 2023; Nie et al., 2023; Shin and Yoon, 2024; Li et al., 2024) introduced dynamic graph structures that adapt to time-varying inputs. Furthermore, Zhang et al. (2025) suggested representing transportation

networks via knowledge graphs, and node embedding methods like LINE (Yao et al., 2018) and Node2vec (Zheng et al., 2020) have been implemented to incorporate spatial information directly into input representations. More recently, a prominent line of research has achieved advanced spatio-temporal modeling by embedding the network topology directly into the attention mechanisms of transformers via Graph Attention Networks (GATs) (Zhang et al., 2019a; Nie et al., 2023; Liu and Meidani, 2024; Hu and Xie, 2025). Zhang et al. (2019a) combined GATs with LSTMs to impose input-dependent variable weights between nodes. Nie et al. (2023) computed speed-pattern-aware adaptive adjacency matrices using graph attention, passing them through graph convolutional layers for network-wide traffic estimation. Additionally, Hu and Xie (2025) utilized an encoder-decoder framework with GATs, proposing both a link-based model to directly predict traffic flow and a path-based model that enforces flow conservation during prediction.

The integration of multi-source datasets has also been emphasized in data-driven traffic modeling approaches. Rempe et al. (2022) and Abdelraouf et al. (2023) utilized floating car data (FCD) and traffic sensor data to estimate network traffic statistics. In Hu et al. (2025), global open multi-source data, including OpenStreetMap (OSM) data and population density maps, were incorporated, along with sensor and FCD data, to develop a universal method for traffic flow estimation. Several studies employed point-of-interest (POI) for the graph representations of transportation networks (Wang et al., 2019; Zhang et al., 2025; Pan et al., 2022). Land-use, congestion, and meteorological information have also been leveraged to enrich the spatio-temporal representations (Yao et al., 2018, 2019; Zhang et al., 2025).

There have been recent efforts to inductively estimate the online traffic states based on partially observable transportation networks (Wu et al., 2021; Abdelraouf et al., 2023; Makridis and Kouvelas, 2023; Nie et al., 2023; Zhou et al., 2025a,b; Hu et al., 2025; Xu et al., 2025). Wu et al. (2021) utilized random subgraph sampling and masking techniques alongside diffusion graph convolutions (Li et al., 2018) to estimate traffic speed at unobserved sensors. Zhou et al. (2025a) integrated the Mixture-of-Experts (MoE) mechanism within transformer architectures into a spatio-temporal graph model for traffic estimation on unobserved nodes. Furthermore, Zhou et al. (2025b) extended the Dirichlet Energy-based Feature Propagation (DEFP) (Rossi et al., 2022) to incorporate directed graphs, thereby addressing the biases inherent in traditional zero-filling inductive GNN methods. Several works (Nie et al., 2023; Abdelraouf et al., 2023; Hu et al., 2025) have developed models that utilize traffic speed data from probe vehicles to estimate traffic flow at unmonitored sensor locations. Nie et al. (2023) addressed the underdetermined and non-equilibrium challenges in traffic flow estimation. By employing a speed pattern-adaptive adjacency matrix and a temporal masked attention mechanism, their proposed model successfully captured non-local spatial dependencies and time-asynchronous correlations in traffic data. Abdelraouf et al. (2023) demonstrated that Graph Convolutional Networks (GCNs) can estimate traffic flow relying solely on speed data, maintaining a reasonable margin of error even at penetration rates of probe vehicles as low as 1.5%. Tang et al. (2022), Ouyang et al. (2024) expand domain adversarial training to spatio-temporal prediction to implement real-time traffic forecasting in sensor-scarce areas. While these studies successfully address the transfer of the spatial representation of online traffic states to sensor-scarce locations utilizing partial historical or real-time observations, our research tackles a fundamentally different challenge. Specifically, we aim to estimate typical, recurring traffic patterns across entirely unobserved networks, relying strictly on static features such as network topology and land-use characteristics without any historical traffic data inputs.

More recently, foundation models have emerged as promising tools for network-wide transportation tasks. He et al. (2025) extracted geospatial embeddings from large language models (LLMs) by designing prompts based on addresses and surrounding places, and evaluated the effectiveness of these embeddings through various downstream tasks, including spatio-temporal traffic forecasting. Nie et al. (2024) proposed a joint estimation framework for delivery demand in observed and unobserved areas by utilizing the method proposed by He et al. (2025). They implemented an end-to-end training approach that used linear probing (Alain and Bengio, 2016) to construct functional graphs based on the LLM-based embeddings among regions of interest and integrated these graphs with a spatio-temporal graph neural network (GNN). Similarly, Balsebre et al. (2024) proposed a pre-training approach for ResNet (He et al., 2016) and BERT (Devlin et al., 2019) using OSM building shapes and relationships, enabling downstream tasks like speed estimation, building function classification, and population density prediction.

Although these recent research efforts have improved the efficiency and accuracy of the network-wide traffic analysis, they primarily focus on dynamic traffic estimation where historical observations in transportation networks are at least partially observable. In other words, the emphasis remains on operational rather than strategic planning contexts. The challenge of traffic estimation and decision support in data-scarce or entirely unobserved regions, where planning decisions are critical, has received limited attention, highlighting the need for further exploration of end-to-end data-driven frameworks for both transductive and inductive traffic estimation.

In addition, while graph convolution and graph attention-based models achieve strong predictive performance, they explicitly require graph message-passing during inference. In contrast, our proposed methodology explores an alternative paradigm: we functionally embed the graph structure into the loss design via graph-contrastive learning. This allows our model to learn highly generalizable representations during training without relying on explicit message-passing in the forward pass. Consequently, this approach not only mitigates the risk of overfitting to specific network topologies but also yields significant computational efficiency.

2.2. Network-wide traffic estimation through traditional travel demand models

Transportation planning has long relied on model-driven frameworks to estimate and manage traffic conditions at the network level. Among these, the trip-based four-step models (FSMs) and activity-based models (ABMs) remain the most widely used. FSMs generate travel demands through four sequential steps: 1) trip generation, 2) trip distribution, 3) mode choice, and 4) assignment. The first three steps generate mode-wise origin-destination (OD) travel demand matrices, and the final step assigns this demand to routes, enabling the estimation of traffic statistics, such as traffic speed, flow, and density.

The trip generation step estimates the number of trips produced and attracted by each region. Traditional methods to do this include regression models (Calabrese et al., 2013; Currans and Clifton, 2018; Justin et al., 2014) and cross-classification models (Justin et al., 2014; Guevara and Thomas, 2007) based on data sources such as household travel surveys, land use, built environment, and socio-demographic data. More recently, machine learning techniques such as artificial neural networks (ANNs) (Lafta and Ismael, 2022; Goel and Sinha, 2008) and GNNs (Liang et al., 2023) have been explored.

The second step, trip distribution, involves distributing each trip to different geographical units, resulting in a pairwise OD matrix. Traditionally, gravity models (Tillema et al., 2006; Abdel-Aal, 2014) have been used frequently for this stage. Recently, new data sources such as mobile phone data (Alexander et al., 2015) and FCD (Gómez et al., 2015) have been incorporated to improve accuracy.

The mode choice splits the OD demand into trips by different travel modes. Random utility modeling approaches, including multinomial (Anas, 1981), nested (Lo et al., 2004), and mixed (Bhat, 2001) logit models, have traditionally been used to model relationships between mode and explanatory variables while accounting for unobserved variables. Recently, machine learning approaches, including ANNs (Lee et al., 2018), tree-based models, and support vector machines (SVMs) (Zhao et al., 2020; Martín-Baos et al., 2023), have shown superior predictive performance compared to conventional logit models.

Finally, the traffic assignment problem (TAP) assigns trips to the network to estimate traffic conditions. Classical equilibrium-based models remain widely used (Bliemer et al., 2017), but their iterative nature makes them computationally expensive, particularly for large-scale networks. Data-driven alternatives, including multi-layer perceptrons (MLPs) (Su et al., 2021), CNNs (Ou et al., 2019; Fan et al., 2023), and GNNs (Hu et al., 2025; Rahman and Hasan, 2023; Liu and Meidani, 2024), have been proposed to address these limitations. Most of these models are trained and evaluated using synthetic data generated from user equilibrium-based simulations based on estimated travel demand inputs (Rahman and Hasan, 2023; Hu and Xie, 2025; Su et al., 2021; Liu and Meidani, 2024; Fan et al., 2023; Ou et al., 2019). While such data allow for controlled experimentation and reproducibility, they may not fully reflect the variability and complexity of real-world traffic dynamics. As a result, models trained and evaluated on synthetic datasets may face challenges in accuracy and generalizability when applied to real-world settings.

ABMs have emerged as an alternative to traditional trip-based FSMs. While FSMs estimate aggregated trip demand at the zonal level, ABMs generate individual tours to capture more realistic travel patterns by modeling interdependencies among daily activities. While the procedural steps in ABMs-activity generation, destination choice, mode choice, and traffic assignment-are analogous to those in FSMs, ABMs generate complete daily individual activity schedules that account for temporal constraints and the sequential relationships between activities (Castiglione et al., 2015).

For activity generation, ABMs frequently employ discrete-continuous econometric models to simultaneously model the type, duration, and timing of activities, effectively capturing the correlations between discrete activity types and their continuous temporal characteristics (Bhat, 2005; Pinjari and Bhat, 2010; Astroza et al., 2018). Recently, machine learning techniques such as K-means clustering (Jiang et al., 2012), random forests (Hafezi et al., 2018), and Variational Autoencoders (VAEs) (Huang et al., 2019; Jeong et al., 2021) have been implemented to learn activity sequences from empirical data.

Mode choice in ABMs typically models complete activity schedules, considering correlations among trips within a tour. Early models often simplified this process by selecting only the primary tour mode (Bowman and Ben-Akiva, 2001; Arentze and Timmermans, 2004), followed by two-stage modeling approaches where trip-level choices were conditioned on the primary mode, OD, and departure time (Bradley et al., 2010; Azevedo et al., 2017). Recent efforts have integrated random utility-based discrete choice models with dynamic programming, enabling the joint estimation of mode choices for all trips within a tour, and thus offering a more realistic representation of travel decisions (Saleem et al., 2018; Hasnine and Habib, 2020).

For traffic assignments, ABMs have primarily focused on integrating agent-based microsimulations (Axhausen et al., 2016; Adnan et al., 2016) with traffic assignment models. While earlier efforts employed static assignment methods (Miller and Roorda, 2003; Goulias et al., 2011), recent studies have emphasized the integration of dynamic traffic assignment (DTA) models to incorporate time-dependent behaviors. These integrated DTA-ABM frameworks can capture dynamic route choices, departure time adjustments, and real-time congestion effects (Heinrichs et al., 2018; Xiong et al., 2021; Agriesti et al., 2023).

Despite the high fidelity and interpretability of traditional travel demand models, both FSMs and ABMs pose challenges for traffic estimations in strategic transportation planning. The multi-step structure of these models often leads to the accumulation of errors across sequential stages. ABMs, while behaviorally richer, lack standardized system architectures and are typically tailored to specific urban regions, making them difficult to calibrate and generalize across different contexts (Miller, 2023). Furthermore, ABMs tend to be computationally intensive, which limits their widespread adoption in practice. Both FSMs and ABMs also struggle with the limited availability of high-quality training data and the complexity of modeling interrelated variables. To address these limitations, this study presents a data-driven, end-to-end framework that leverages recent advancements in machine learning for both transductive and inductive traffic estimations. This framework is based solely on network topology and land-use data to estimate traffic in the absence of historical observations.

3. Definitions and problem statement

In this section, we explain the types of datasets used in this study and formally define the transductive and inductive traffic estimation problems. The goal of both problems is to estimate traffic statistics solely based on network structures and standardized land-use information for road segments that were not observed during the training phase. An overview of the proposed GTE-GC framework is presented in Fig. 1.

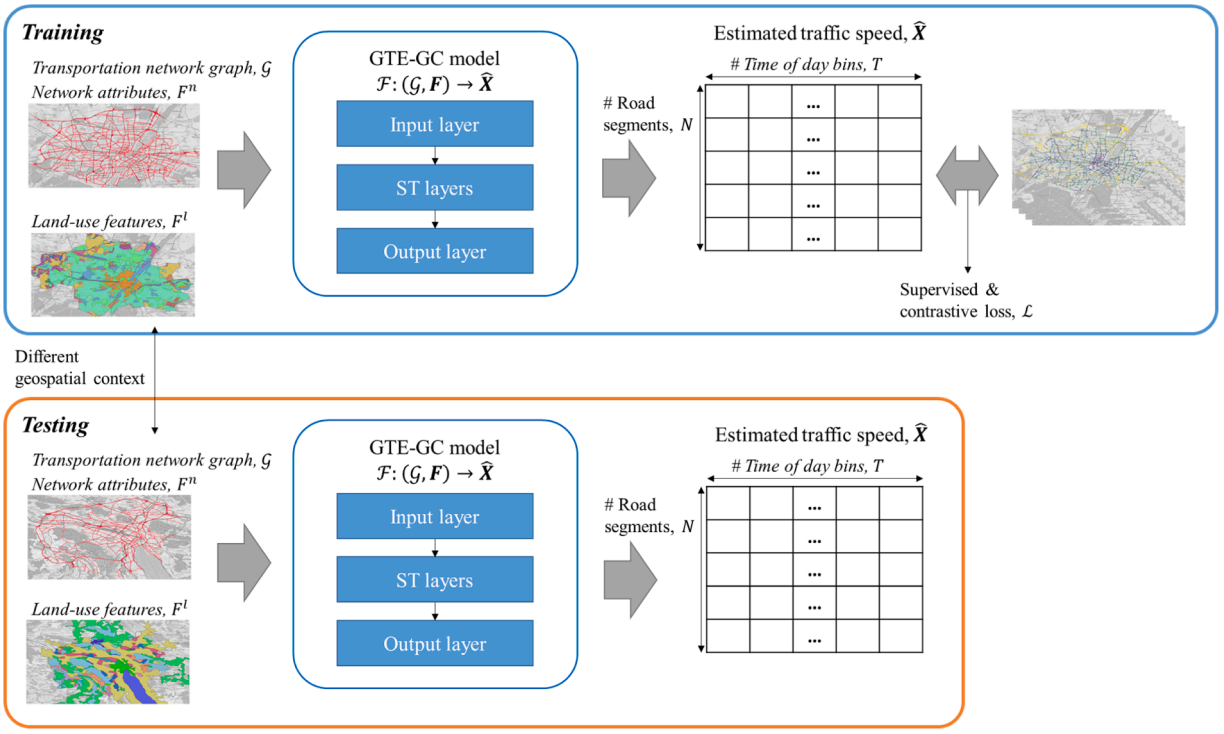


Fig. 1. Overview of GTE-GC. The model takes transportation network data and land-use data as input and is trained for traffic estimation. The model is then validated and tested under unobserved environments where no historical traffic data is available to fine-tune the model parameters.

3.1. Data description

This study utilizes datasets of three different structures: traffic speed data, transportation network data, and land-use data, which are detailed as follows:

Traffic speed data. The proposed framework utilizes traffic speed data as the target signal for estimation. Although traffic flow may be more commonly used to evaluate transportation network performance, traffic speed is also a critical metric with wide-ranging applications, including infrastructure planning (Narayanaswami, 2017), travel time estimation (Zhang et al., 2019b), and accident hotspot identification (Sajed et al., 2019).

Traffic speed data typically provides link-level speed measurements at intervals ranging from 5 min to an hour. These data are primarily collected through stationary traffic sensors or probe vehicles. Stationary sensors, such as loop detectors, cameras, RFID, or LiDAR, record vehicle speeds at fixed locations. In contrast, FCD from probe vehicles captures positions and speeds, which are then mapped to road segments to compute average traffic statistics over defined time windows. Example datasets include HKeMobility,[1] PeMS,[2] and TomTom Move.[3]

Transportation network data. The transportation network data comprises diverse spatial and structural information about road networks, including geographical location, speed limit, number of lanes, length of the road segment, and road type. In many existing studies, it has been shown that the inclusion of such data contributes to more accurate network-wide traffic analyses (Loder et al., 2019; Hu et al., 2025; Liu and Meidani, 2024; Shin and Yoon, 2022). Example datasets include TomTom Move[4] and OpenStreetMap.[5]

Land-use data. Since the 1950s, numerous studies have focused on exploring the relationship between land-use and urban traffic, demonstrating that incorporating land-use into the modeling framework can help address transportation problems (Mitchell and Rapkin, 1954; Badoe and Miller, 2000; Zhang et al., 2025). While land-use categories vary across nations and institutions, several data sources cover larger regions with consistency, such as OpenStreetMap (See footnote 4) and Copernicus.[6]

[1] https://www.hkemobility.gov.hk/
[2] https://dot.ca.gov/programs/traffic-operations/mpr/pems-source
[3] https://move.tomtom.com
[4] https://move.tomtom.com
[5] https://www.openstreetmap.org
[6] https://land.copernicus.eu/en

Detailed descriptions of the datasets used in this study are presented in Section 5.1.

3.2. Traffic estimation problems

Definition 1. Transportation network graph. Let $\mathcal{G} = (\boldsymbol{V}, \boldsymbol{E}, \boldsymbol{F}, \boldsymbol{A})$ denote a transportation network, where $\boldsymbol{V}$ is the set of road segments (nodes) with size $|\boldsymbol{V}| = N$, and $\boldsymbol{E}$ is the set of edges representing their connectivity. $\boldsymbol{A} \in \mathbb{R}^{N \times N}$ denotes the adjacency matrix of the network, where $a_{ij} = 1$ if an edge exists between v_i and v_j, and 0 otherwise. Each node $v_i \in \boldsymbol{V}$ is associated with a feature vector $\boldsymbol{f}_i \in \mathbb{R}^{d_f}$ that encodes information from two sources:

- Network attributes, $\boldsymbol{f}_i^n \in \mathbb{R}^{d_n}$: structural properties of road segments such as segment length, functional road class, speed limit, the number of nearby segments, and centrality measures.
- Land-Use features: spatial features representing land-use composition around the segment, encoded as a vector $\boldsymbol{f}_i^l \in \mathbb{R}^{d_l}$, where d_l is the number of aggregated land-use categories.

We denote the concatenated static feature matrix as $\boldsymbol{F} = [\{\boldsymbol{f}_1^n, \ldots, \boldsymbol{f}_N^n\}, \{\boldsymbol{f}_i^l, \ldots, \boldsymbol{f}_N^l\}] \in \mathbb{R}^{N \times d_f}$, where $d_f = d_n + d_l$ is the total dimension of the combined network and land-use features.

Definition 2. Speed data. Let $\boldsymbol{X} \in \mathbb{R}^{N \times T}$ denote the traffic speed values associated with a transportation network graph, where $x_{n,t}$ represents the average hourly traffic speed on weekdays on road segment v_i for a given time-of-day t. Note that $\boldsymbol{X}$ is the hourly traffic speed averaged over all weekdays. This formulation aims to estimate the hourly traffic pattern on typical weekdays for a target (sub)network using learned relationships from another (sub)network.

Problem formulation. Given a transportation network graph $\mathcal{G} = (\boldsymbol{V}, \boldsymbol{E}, \boldsymbol{F}, \boldsymbol{A})$ and the associated traffic speed matrix $\boldsymbol{X} \in \mathbb{R}^{N \times T}$, the objective is to learn a function $\mathcal{F}$ such that

$$\mathcal{F} : (\mathcal{G}, \boldsymbol{F}) \rightarrow \hat{\boldsymbol{X}}, \tag{1}$$

that estimates the hourly traffic speed profiles $\hat{\boldsymbol{X}}$ across the network. The model parameters are trained to minimize a loss function $\mathcal{L}(\hat{\boldsymbol{X}}, \boldsymbol{X})$ measuring the difference between estimated and true speed values. Note that the loss is calculated for the time and nodes where the speed is available, although we retain all nodes in the transportation networks, including those that lack speed information.

We define two learning settings:

- *Transductive Traffic Estimation*: The model is trained and evaluated on the same transportation network graph $\mathcal{G}$. A subset of nodes $\boldsymbol{V}_{train} \subset \boldsymbol{V}$ is used for training, and the remaining nodes $\boldsymbol{V}_{test} = \boldsymbol{V} \setminus \boldsymbol{V}_{\text{train}}$ are used for evaluation. The other elements of the graph are also divided as $\mathcal{G}_{train} = (\boldsymbol{V}_{train}, \boldsymbol{E}_{train}, \boldsymbol{F}_{train}, \boldsymbol{A}_{train})$ and $\mathcal{G}_{test} = (\boldsymbol{V}_{test}, \boldsymbol{E}_{test}, \boldsymbol{F}_{test}, \boldsymbol{A}_{test})$ based on the train and test node subsets. In this setting, the train and test subnetworks are within the same geospatial context.
- *Inductive Traffic Estimation*: The model is trained on source network graphs $\mathcal{G}^{(s)} = (\boldsymbol{V}^{(s)}, \boldsymbol{E}^{(s)}, \boldsymbol{F}^{(s)}, \boldsymbol{A}^{(s)})$ with corresponding traffic speed data $\boldsymbol{X}^{(s)}$, and evaluated on a completely unseen target network $\mathcal{G}^{(t)} = (\boldsymbol{V}^{(t)}, \boldsymbol{E}^{(t)}, \boldsymbol{F}^{(t)}, \boldsymbol{A}^{(t)})$. Unlike the transductive task, a full network rather than a sub-network is used for training. The goal is to estimate the hourly average traffic speed matrix $\boldsymbol{X}^{(t)}$ using only $\boldsymbol{F}^{(t)}$ and $\boldsymbol{A}^{(t)}$, without access to any speed labels from the target graph during training. This setup reflects the scenarios where historical traffic data is unavailable for the target city or region.

While both tasks address scenarios where historical traffic speed profiles are unavailable in the test regions, they fundamentally differ in the scale of spatial extrapolation and the magnitude of the underlying distribution shift. For the transductive task, the city network is partitioned into spatially disjoint clusters (regions) for training and testing. This spatial segmentation is intentionally designed to simulate regional sensor absence, such as in newly developed urban expansion zones or peripheral suburban areas where sensor deployment is structurally limited. This spatial segmentation setting presents a more challenging task than random scattered masking, since it forces the model to generalize to entirely unobserved sub-networks without relying on the localized spatial autocorrelation provided by closely neighboring observed segments. Although this regional partitioning inherently introduces a spatial distribution shift between the training and testing sub-networks, the overall macroscopic traffic distribution of the city remains partially known. In contrast, the inductive traffic estimation task requires estimating traffic in an entirely unseen target city. Consequently, the distribution shift experienced in the inductive setting is far greater, as the model must generalize across completely different urban forms, network topologies, and travel behaviors without any prior knowledge of the target network's traffic distribution.

It is also important to explicitly define the scope of the target variable within this framework. Our formulation specifically targets the estimation of typical, recurring weekday traffic profiles by utilizing hourly speeds averaged over multiple weekdays. The objective of our study is to estimate the long-term recurring traffic patterns that are primarily governed by static road-network topology and urban land-use distributions. Note that these two tasks are trained independently on separate datasets. Although they share the same objective function and model architecture, they do not share the same weights.

4. Method

In this section, we explain the implementation of the GTE-GC (Generalizable Traffic Estimation based on Graph Contrastive learning) framework on the transductive and inductive traffic estimation tasks. We first explain the input layer and the temporal encoding layer. Then, we introduce a spatio-temporal layer (ST-layer) that consists of a self-attention layer, a spatial feed-forward neural network (FFNN), and a feature fusion layer. Finally, the graph contrastive learning objective for inductive estimation is explained. The overview of the suggested model is presented in Fig. 2.

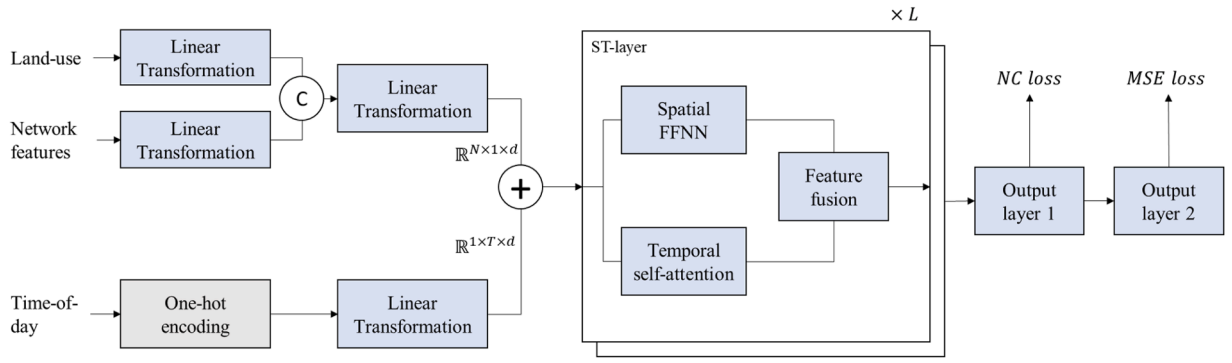


Fig. 2. Overall model architecture for GTE-GC.

4.1. Input layer

The input layer incorporates two datasets-(1) transportation network data and (2) land-use data-to construct spatial representations, and time-of-day to capture temporal correlations.

To enable the model to capture the contribution of each input dataset, we first apply separate linear transformations to the network and land-use features. For network attributes, $\boldsymbol{f}_i^n \in \mathbb{R}^{d_n}$, and land-use features, $\boldsymbol{f}_i^l \in \mathbb{R}^{d_l}$, of node v_i, linear layers are implemented to extract hidden representations $\boldsymbol{h}_i^n \in \mathbb{R}^{d_1}$, and $\boldsymbol{h}_i^l \in \mathbb{R}^{d_2}$:

$$\boldsymbol{h}_i^n = \boldsymbol{f}_i^n \boldsymbol{W}_n + \boldsymbol{b}_n, \quad \boldsymbol{h}_i^l = \boldsymbol{f}_i^l \boldsymbol{W}_l + \boldsymbol{b}_l, \tag{2}$$

where $\boldsymbol{W}_n, \boldsymbol{W}_l, \boldsymbol{b}_n, \boldsymbol{b}_l$ are learnable parameters. These representations are concatenated and passed through a non-linear ReLU activation, batch normalization, and dropout to stabilize and regularize learning:

$$\boldsymbol{h}_i^{concat} = \mathrm{Dropout}\big(\mathrm{BatchNorm}\big(\mathrm{ReLU}\big(\big[\boldsymbol{h}_i^n \| \boldsymbol{h}_i^l\big]\big)\big)\big), \tag{3}$$

where $[\cdot\|\cdot]$ denotes vector concatenation. Then, $\boldsymbol{h}_i^{concat}$ is passed through another transformation to obtain a joint spatial representation $\boldsymbol{hs}_i^{(0)} \in \mathbb{R}^d$ for node v_i:

$$\boldsymbol{hs}_i^{(0)} = \boldsymbol{h}_i^{concat} \boldsymbol{W}_{concat} + \boldsymbol{b}_{concat}, \tag{4}$$

where $\boldsymbol{W}_{concat}$ and $\boldsymbol{b}_{concat}$ are learnable parameters. This enables the model to consider contributions of network and land-use features independently, allowing it to learn their distinct contributions before combining them into a joint spatial representation.

The framework utilizes one-hot encoding to represent the temporal information. Each time-of-day slot is represented as a binary vector of length T. The one-hot encoding treats each time slot as a distinct categorical entity without implying any ordinal or metric relationship between them. This can help inductive traffic estimation, where the context of relationship between consecutive time-of-day bins may differ among regions. The vanilla one-hot encoding allows the model to learn temporal correlations without enforcing artificial continuity or distance assumptions between time-of-days. For temporal representation, the encoded time-of-day matrix $\boldsymbol{E} \in \mathbb{R}^{T\times T}$ is passed through a linear transformation to produce the temporal representation $\boldsymbol{Ht}^{(0)} \in \mathbb{R}^d$:

$$\boldsymbol{Ht}^{(0)} = \boldsymbol{E}\boldsymbol{W}_t + \boldsymbol{b}_t, \tag{5}$$

where $\boldsymbol{W}_t$ and $\boldsymbol{b}_t$ are learnable parameters.

Finally, the spatial representation, $\boldsymbol{Hs}^{(0)}$, and temporal representation, $\boldsymbol{Ht}^{(0)}$, are aggregated through an addition to generate the spatio-temporal representation, $\boldsymbol{H}^{(0)} \in \mathbb{R}^{N\times T\times d}$, at the input layer:

$$\boldsymbol{H}^{(0)} = \mathrm{BatchNorm}\big(\mathrm{ReLU}\big(\boldsymbol{Hs}^{(0)}.unsqueeze(1) + \boldsymbol{Ht}^{(0)}.unsqueeze(0)\big)\big), \tag{6}$$

where $\boldsymbol{M}.unsqueez(i)$ indicates expansion of tensor $\boldsymbol{M}$ along ith axis.

4.2. Spatio-temporal layer

The ST-layer consists of three components: a spatial FFNN, a temporal feature extractor, and a feature fusion module. At each lth ST-layer, encoders learn latent representations from the input tensor $\boldsymbol{H}^{(l-1)} \in \mathbb{R}^{N\times T\times d}$, capturing road network features and time-of-day variations, respectively. These representations are then integrated through the feature fusion layer to produce a joint spatio-temporal representation.

4.2.1. Spatial feed-forward neural networks

In this layer, we apply an FFNN to extract spatial representations for each road segment. Let $\boldsymbol{H}^{(l)} \in \mathbb{R}^{N\times T\times d}$ denote the input spatio-temporal tensor for lth ST-layer. For each node v_i and time-of-day t, we apply a linear transformation across nodes to generate spatial representation $\boldsymbol{Hs}_{i,t}^{(l)} \in \mathbb{R}^d$:

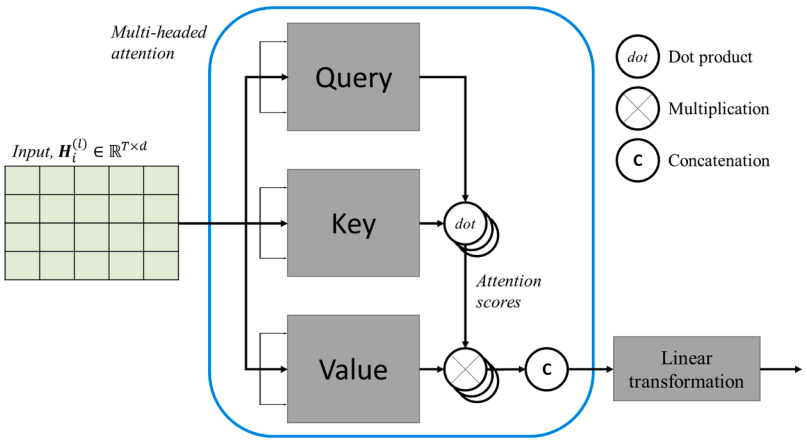


Fig. 3. Illustration of temporal self-attention mechanism. Query, key, and value indicate intermediate feed-forward layers of the multi-headed self-attention mechanism in the temporal layer.

$$\boldsymbol{Hs}_{i,t}^{(l)} = \boldsymbol{H}_{i,t}^{(l-1)}\boldsymbol{W}_s + \boldsymbol{b}_s, \tag{7}$$

where $\boldsymbol{W}_s$ and $\boldsymbol{b}_s$ are learnable parameters. $\boldsymbol{Hs}^{(l)} \in \mathbb{R}^{N\times T\times d}$ is the spatial representation at lth ST layer.

Although no explicit message passing is used in this layer, the model can learn spatial patterns from the input network attributes that are processed to implicitly capture topological context (e.g., the number of nearby segments and centrality measures). Moreover, the entire network topology is explicitly leveraged later in the training process through the graph contrastive learning objective.

4.2.2. Temporal feature extraction

To capture hour-specific traffic patterns and temporal correlations, we apply a self-attention mechanism along the temporal dimension. This sub-layer enables the model to flexibly weigh and contextualize traffic speed patterns on each time-of-day relative to others.

Given the input spatio-temporal representation $\boldsymbol{H}^{(l-1)} \in \mathbb{R}^{N\times T\times d}$ for lth ST-layer, we apply the self-attention mechanism (Vaswani et al., 2017) along the temporal dimension with T time steps. Formally, we first compute the similarity score $s_{t_1,t_2}^{(l)}$ between time-of-day t_1 and t_2 for any node v_i:

$$s_{t_1,t_2}^{(l)} = \frac{(\boldsymbol{H}_{i,t_1}^{(l-1)}\boldsymbol{W}_q^{(l)} + \boldsymbol{b}_q^{(l)}) \cdot (\boldsymbol{H}_{i,t_2}^{(l-1)}\boldsymbol{W}_k^{(l)} + \boldsymbol{b}_k^{(l)})}{\sqrt{d}}, \tag{8}$$

where $\boldsymbol{W}_q^{(l)}, \boldsymbol{W}_k^{(l)}, \boldsymbol{b}_q^{(l)}, \boldsymbol{b}_k^{(l)}$ are learnable parameters and $\cdot$ is the dot product operation. Then, we apply the softmax function to the similarity score to get the attention score, $\alpha_{t_1,t_2}^{(l)}$:

$$\alpha_{t_1,t_2}^{(l)} = \frac{\exp\left(s_{t_1,t_2}^{(l)}\right)}{\sum_{t=1}^{T}\exp\left(s_{t_1,t}^{(l)}\right)}. \tag{9}$$

Then, we extract the temporal representation $\boldsymbol{Ht}_{i,t}^{\prime(l)}$ for node v_i and time-of-day t:

$$\boldsymbol{Ht}_{i,t}^{\prime(l)} = \sum_{t=1}^{T}\alpha_{t_1,t}\left(\boldsymbol{H}_{i,t}^{(l-1)}\boldsymbol{W}_v^{(l)} + \boldsymbol{b}_v^{(l)}\right), \tag{10}$$

where $\boldsymbol{W}_v^{(l)}, \boldsymbol{b}_v^{(l)}$ are learnable parameters. Here, we adopt the multi-head attention mechanism (Vaswani et al., 2017) which concatenates temporal representation from K heads to generate final temporal representation:

$$\boldsymbol{Ht}_{i,t}^{\prime(l)} = \mathrm{Concat}_{k=1}^{K}\left(\boldsymbol{Ht}_{i,t}^{\prime(k,l-1)}\right). \tag{11}$$

For the multi-head mechanism, the hidden dimensions, d, in Eqs. (8)–(10) are d/K. Finally, we produce the final temporal representation, $\boldsymbol{Ht}_{i,t}^{(l)}$, for node v_i and time-of-day, t, through a feed-forward layer:

$$\boldsymbol{Ht}_{i,t}^{(l)} = \boldsymbol{Ht}_{i,t}^{\prime(l)}\boldsymbol{W}_{ta} + \boldsymbol{b}_{ta}, \tag{12}$$

where $\boldsymbol{W}_{ta}$ and $\boldsymbol{b}_{ta}$ are learnable parameters. Fig. 3 illustrates the temporal attention mechanism.

4.2.3. Feature fusion

After extracting spatial and temporal features independently, we integrate them using a feature fusion mechanism introduced in Zheng et al. (2020). This layer adaptively combines the outputs of the spatial FFNN and the temporal self-attention module by learning a gating function that controls the relative influence of each representation.

Let $\boldsymbol{Hs}^{(l)} \in \mathbb{R}^{N\times T\times d}$ and $\boldsymbol{Ht}^{(l)} \in \mathbb{R}^{N\times T\times d}$ denote the spatial and temporal feature tensors at the lth ST layer, respectively. For each node v_i and time-of-day t, we compute the fused representation:

$$\boldsymbol{H}'^{(l)} = z \odot \boldsymbol{Hs}^{(l)} + (1 - z) \odot \boldsymbol{Ht}^{(l)}, \tag{13}$$

where $\odot$ indicates element-wise multiplication, and $z_{i,t} \in \mathbb{R}^d$ is the gating vector:

$$z_{i,t} = \sigma\left(\boldsymbol{Hs}^{(l)}\boldsymbol{W}_{zs} + \boldsymbol{Ht}^{(l)}\boldsymbol{W}_{zt} + \boldsymbol{b}_z\right), \tag{14}$$

where $\boldsymbol{W}_{zs}$, $\boldsymbol{W}_{zt} \in \mathbb{R}^{d\times d}$, and $\boldsymbol{b}_z \in \mathbb{R}^d$ are learnable parameters, and $\sigma(\cdot)$ is the sigmoid activation function. Finally, the output of the lth ST layer $\boldsymbol{H}^{(l)}$ is obtained through two additional feed-forward layers, batch normalization, and non-linear activation for the output, $\boldsymbol{H}^{(l)}_{i,t} \in \mathbb{R}^{N\times T\times d}$:

$$\boldsymbol{H}^{(l)}_{i,t} = \mathrm{BatchNorm}\left(\mathrm{ReLU}\left(\boldsymbol{H}'^{(l)}_{i,t}\boldsymbol{W}_{f1} + \boldsymbol{b}_{f1}\right)\right)\boldsymbol{W}_{f2} + \boldsymbol{b}_{f2}. \tag{15}$$

4.3. Output layer and learning objectives

4.3.1. Output layer

After L ST-layers, the final spatio-temporal representation $\boldsymbol{H}^{(L)} \in \mathbb{R}^{N\times T\times d}$ is passed through two feed-forward layers to produce the output:

$$\boldsymbol{X}'_{i,t} = \mathrm{BatchNorm}\left(\mathrm{ReLU}\left(\boldsymbol{H}^{(L)}_{i,t}\boldsymbol{W}_{o1} + \boldsymbol{b}_{o1}\right)\right), \quad \hat{\boldsymbol{X}}_{i,t} = \boldsymbol{X}'_{i,t}\boldsymbol{W}_{o2} + \boldsymbol{b}_{o2}, \tag{16}$$

where $\boldsymbol{W}_{o1} \in \mathbb{R}^{d\times d}$, $\boldsymbol{W}_{o2} \in \mathbb{R}^{d\times 1}$ are learnable weights and $\boldsymbol{b}_{o1}, \boldsymbol{b}_{o2}$ are corresponding biases.

4.3.2. Learning objectives

To enhance the model's ability to generalize across transportation networks with different geospatial contexts, we adopt a contrastive learning objective (Chen et al., 2020; You et al., 2020; Hu et al., 2021). Contrastive learning is a self-supervised learning framework that trains models to learn semantic representations by narrowing down the distance between positive (similar) samples and widening it between negative (dissimilar) ones in the latent space. In the broader machine learning and graph representation literature, contrastive objectives are widely recognized for their effectiveness in transfer learning tasks and their robustness to data perturbations and distribution shifts. To incorporate the transportation network graph while retaining these benefits, we use a neighboring contrastive learning strategy inspired by Graph-MLP (Hu et al., 2021). By enforcing topological consistency in the loss function rather than the forward pass, this contrastive objective helps the model learn generalizable, structure-aware, and invariant embeddings in a computationally efficient manner. Consequently, the learned representations are significantly less dependent on specific geographic contexts compared to standard message-passing GNNs, which is particularly crucial for maintaining high performance in inductive traffic estimation.

Let the mean of intermediate output representation be denoted $\bar{\boldsymbol{X}}' = \mathrm{mean}(\boldsymbol{X}', dim = 1) \in \mathbb{R}^{N\times d}$. For each node v_i, we define positive samples as the r-hop topological neighbors of node v_i, while the rest are defined as negative samples.

The neighboring contrastive loss for node v_i is given by:

$$L_{NC,i} = -\log \frac{\sum_{j\in\mathcal{P}_i} \gamma_{ij} \exp\left(\mathrm{sim}(\bar{\boldsymbol{X}}_i, \bar{\boldsymbol{X}}_j)/\tau\right)}{\sum_{k\in N} \exp\left(\mathrm{sim}(\bar{\boldsymbol{X}}_i, \bar{\boldsymbol{X}}_k)/\tau\right)}, \tag{17}$$

where the neighbor weights γ_{ij} can be defined:

$$\gamma_{ij} = \bar{\boldsymbol{A}}^r_{ij}, \quad \bar{\boldsymbol{A}} = \bar{\boldsymbol{D}}^{-1}\bar{\boldsymbol{A}}, \quad \bar{\boldsymbol{A}} = \boldsymbol{A} + \boldsymbol{I}, \tag{18}$$

τ is a temperature parameter, $\mathcal{P}_i$ is the set of positive nodes (neighbors), and γ_{ij} is a weighting coefficient based on adjacency proximity. The similarity $\mathrm{sim}(\cdot,\cdot)$ denotes cosine similarity, which can be defined for arbitrary feature vectors $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$:

$$\mathrm{sim}(\boldsymbol{x}_1, \boldsymbol{x}_2) = \frac{\boldsymbol{x}_1 \cdot \boldsymbol{x}_2}{\|\boldsymbol{x}_1\|\|\boldsymbol{x}_2\|}. \tag{19}$$

This loss encourages representations of topologically close segments to lie near each other in the embedding space while distancing those that are unrelated. Importantly, this training strategy helps the model learn latent features based on topological structures of transportation networks, making it well-suited for inductive traffic estimation, where the traffic speed of the target network is to be estimated based on topological structures and geospatial contexts without historical traffic data.

Alongside the contrastive objective, we use a standard mean squared error (MSE) loss to supervise the estimation of traffic speed:

$$L_{MSE} = \frac{1}{N^{obs}T} \sum_{v_i\in\boldsymbol{V}^{obs}} \sum_{t=1}^{T} \left(\hat{x}_{i,t} - x_{i,t}\right)^2, \tag{20}$$

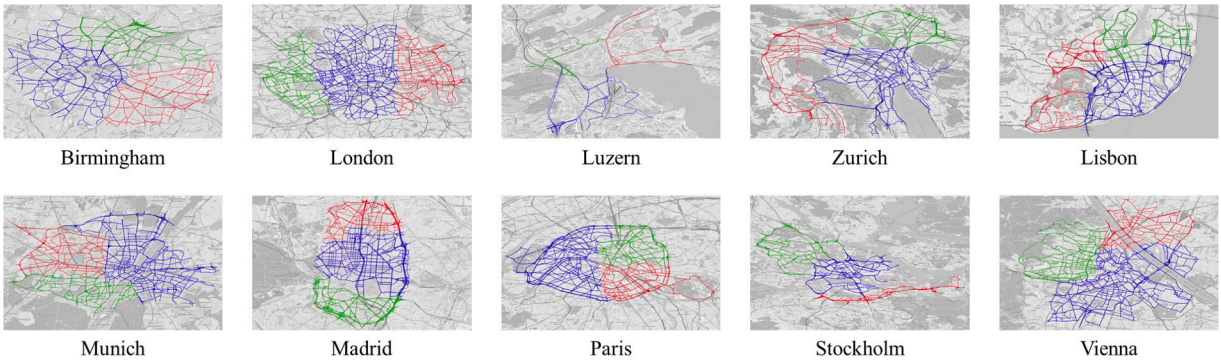


Fig. 4. Transportation networks of study areas. The colors of segments indicate subnetworks used for the training (blue), validation (green), and testing (red). The training subnetworks include 50–60% of the total road segments, and the validation and testing subnetworks, 20–27% in all cities.

where $\boldsymbol{V}^{obs}$ denotes the set of observed nodes used for training, $N^{obs} = |\boldsymbol{V}^{obs}|$ is the number of observed nodes, and $x_{i,t}, \hat{x}_{i,t}$ are the ground-truth and estimated traffic speed values.

The final objective combines both losses:

$$\mathcal{L} = L_{MSE} + \alpha \sum_{v_i \in \boldsymbol{V}^{obs}} L_{NC,i}, \tag{21}$$

where α is a hyperparameter controlling the contribution of the contrastive loss. This joint objective enables the model to improve both estimation accuracy and structural awareness while maintaining inductive generalizability to unobserved networks.

By leveraging this contrastive objective, the model successfully captures the underlying network topology without relying on explicit layer-wise message passing in the forward pass, significantly improving the overall computational efficiency. Technically, in a standard graph convolutional layer, the topological aggregation results in a time complexity of $\mathcal{O}(|E|d + Nd^2)$. In contrast, our proposed formulation decouples the graph structure during the forward pass, effectively reducing the complexity to $\mathcal{O}(Nd^2)$. As demonstrated in the subsequent experiments, this architecture achieves comparable or even superior traffic estimation performance while maintaining computational efficiency.

5. Experiments

In this section, we present the experiments of our study across ten European cities. Section 5.1 presents the detailed description of the datasets used in this study. Section 5.2 explains the comparison models, experimental settings, evaluation metrics, and hyperparameter settings. The results and analyses are presented from Sections 5.3 to 5.7.

5.1. Datasets

The proposed framework is evaluated using the datasets from 10 European cities: Birmingham, London, Luzern, Zurich, Paris, Munich, Lisbon, Stockholm, Madrid, and Vienna. The transportation network of each city is shown in Fig. 4.

5.1.1. Traffic speed data and transportation network data

The traffic speed and transportation network data used in this study were obtained from TomTom,[7] a company specializing in navigation devices and location-based services. TomTom collects data from traveling vehicles that use their hardware devices or mapping applications. TomTom TrafficStats provides various statistics, including the number of samples, mean and median travel times, mean and median speeds, harmonic mean speed, speed standard deviation, and speed percentiles for each segment and 15-min interval. In this study, we use the mean speed as the target variable for estimation. We collected data from Monday, November 9, to Friday, November 13, 2021, covering all road segments at 15-min intervals between 5:00 and 23:00. This period was selected to minimize the impact of the COVID-19 pandemic, mobility restrictions, main holidays, and extreme weather events. The mean speeds were then aggregated to an hourly resolution and averaged over five weekdays, resulting in a data matrix of size $\boldsymbol{X} \in \mathbb{R}^{N \times 18}$. Fig. 5 shows the city-wide average diurnal hourly traffic speeds from Monday to Friday, alongside the overall five-day average for all cities. Note that these speed profiles are aggregated to show a single representative temporal pattern per city, which facilitates macro-level comparisons.

The transportation network data consists of the structural attributes of each road segment, including the segment ID, length, functional road class (FRC), speed limit, and geographic location. The FRC ranges from 0 to 5, depending on the functional hierarchy of the road segments, where 0 represents the motorways and 5 represents the major local roads. Please note that smaller minor roads

[7] https://move.tomtom.com

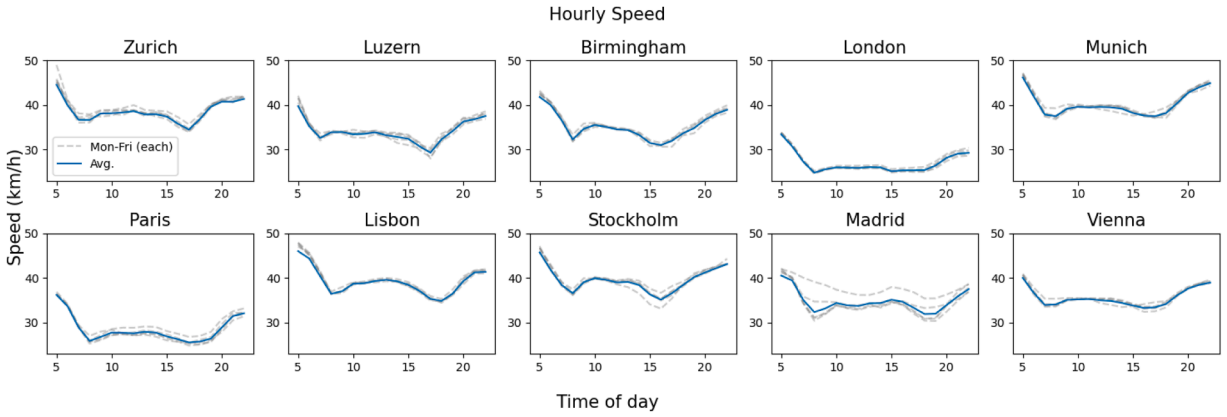


Fig. 5. Hourly speed profiles across different transportation networks. Each of the five gray dashed lines represents the network-wide average hourly speed for a specific weekday (Monday through Friday). The solid blue line represents the overall average traffic speed across all five weekdays. Note that spatial averaging is used here strictly to visualize macro-level daily trends and inter-city differences.

Table 1
Number of road segments and traffic speed statistics in each study city.

City	# Roads for training	# Roads for validation	# Roads for testing	Speed Limit (km/h)	Speed (km/h)
Zurich, CHE	8079	2691	2751	53.83 ± 16.06	37.60 ± 16.06
Luzern, CHE	680	288	324	48.96 ± 15.52	33.18 ± 12.79
Birmingham, GBR	6150	2842	2914	46.45 ± 11.27	34.38 ± 10.65
London, GBR	21,408	8011	9186	36.08 ± 7.48	26.53 ± 7.20
Munich, DEU	15,416	5438	6124	51.37 ± 18.62	39.14 ± 16.15
Paris, FRA	11,151	4307	5546	35.78 ± 11.63	27.87 ± 10.87
Lisbon, PRT	7025	3011	3733	51.04 ± 14.31	38.04 ± 15.88
Stockholm, SWE	7691	3850	3106	49.23 ± 14.08	38.55 ± 17.12
Madrid, ESP	7360	3609	3150	49.98 ± 9.57	33.93 ± 14.30
Vienna, AUT	13,715	5443	4910	48.00 ± 12.25	34.89 ± 12.51

(i.e., FRC 6 and 7) were excluded from our analysis. Due to the limited penetration rate of probe vehicles on these lower-order streets, their resulting speed observations tend to be highly unstable and noisy, with a substantial proportion of missing values. In addition to the original features, we processed and included the number of nearby segments to impose spatial context directly on the dataset. Specifically, for each road segment v_i, the number of nearby segments, $NSegment(v_i)$, is defined as the number of other road segments that intersect with the buffer polygon circle, $B(v_i, \beta_s)$, surrounding v_i with β_s radius:

$$NSegment(v_i) = \sum_j s_j,$$
$$s_j = \begin{cases} 1, & \text{if } \left(v_j \cap B(v_i, \beta_s)\right) \neq \emptyset \\ 0, & \text{otherwise} \end{cases} \quad \forall v_{j \neq i} \in \boldsymbol{V} \tag{22}$$

The inclusion of this explicitly engineered spatial feature provides supplementary information that cannot be captured by standard graph representations. While graph representations effectively encode topological connectivity (i.e., how segments are linked through junctions), they inherently abstract away the physical proximity and spatial scale of the network in Euclidean space. For instance, two road segments might possess the exact same topological degree (number of connected edges) but exist in vastly different geographic contexts-one in a highly dense urban grid and the other in a sparse suburban area. By explicitly providing $NSegment(v_i)$ to the spatial encoder, we equip the forward model with vital cues about the local Euclidean density of the transportation network.

Table 1 presents the number of segments for training, validation, and testing sets along with speed statistics of each transportation network.

5.1.2. Land-use data

For land-use data, we employed the CORINE Land Cover (CLC) 2018 dataset,[8] a standardized and harmonized land cover dataset developed under the Coordination of Information on the Environment (CORINE) program by the European Commission. CLC includes five broad level 1 land cover classes, and more detailed level 2 and level 3 classes. While the original classification comprises 44

[8] https://land.copernicus.eu/en/products/corine-land-cover/clc2018#download

Table 2
Corine land cover 2018 category codes and descriptions.

Code	Description	Code	Description
111	Continuous urban fabric	133	Construction sites
112	Discontinuous urban fabric	141	Green urban areas
121	Industrial or commercial units	142	Sports and leisure facilities
122	Road and rail networks and associated land	200	Agricultural lands
123	Port areas	300	Forest and seminatural areas
124	Airports	400	Wetlands
131	Mineral extraction sites	500	Water bodies
132	Dump sites		

level 3 classes, we aggregated them into 15 classes to reduce sparsity in the input features. The aggregated land cover codes and descriptions are introduced in Table 2.

The CLC dataset provides land-use codes at the NUTS (Nomenclature of Territorial Units for Statistics) region level, which do not directly correspond to individual road segments. To spatially align the land-use data with the transportation network, we created a buffer around each road segment. We calculated the proportion of the buffer area that overlaps with each land-use category at the NUTS-3 level. The land-use information for each road segment v_i was then encoded into a feature vector $\boldsymbol{f}_i^l \in \mathbb{R}^{d_l}$:

$$\boldsymbol{f}_i^l = \left[f_{i,1}^l, f_{i,2}^l, \dots, f_{i,u}^l, \dots f_{i,d_l}^l\right] \forall u = 1, 2, \dots, d_l, \text{and}$$
$$f_{i,u}^l = \frac{Area\left(B(v_i, \beta_f) \cap L_u\right)}{Area\left(B(v_i, \beta_f)\right)}, \tag{23}$$

where d_l is the number of land-use categories, $f_{i,u}^l$ is the fraction of the buffer space $B(v_i, \beta_f)$ that falls in the uth land-use category L_u, and $Area(\cdot)$ is the area of the space inside the parenthesis.

To empirically verify the spatial dependence of the transportation network and land use features, we computed Moran's I (Cliff and Ord, 1981), for the input features across the studied networks. The results demonstrated exceptionally strong positive spatial autocorrelation, with many topology-derived and contextual descriptors exhibiting Moran's I values exceeding 0.90. This inherent spatial correlation ensures that our framework is capable of learning topology-aware representations and achieving strong generalizability without relying on explicit message passing during the forward pass.

5.2. Baselines and experimental settings

For the evaluation, we compare the traffic estimation results of the proposed model with the following baseline models:

- *Graph Convolutional Networks (GCN)* implements a 2-layer GCN (Kipf and Welling, 2017) architecture attached to the suggested input encoder. This model only considers spatial correlations.
- *GTE w/o TE* adopts a linear transformation to expand the temporal dimension from the input features instead of time-of-day encodings.
- *GTE-DAT (Domain Adversarial Training)* integrates a domain adversarial training framework into the base GTE model. It employs a gradient reversal layer and a domain discriminator to align the feature representations between the source and target networks. This baseline is exclusively implemented for the inductive traffic estimation task.
- *GTE* trains GTE-GC without graph contrastive loss (Eq. (17)). The learning objective is solely based on the supervised loss (Eq. (20)).
- *GTE-GCN* replaces the spatial FFNN in GTE with GCN (Kipf and Welling, 2017).
- *GTE-GCN-GC* replaces the spatial FFNN in GTE with GCN (Kipf and Welling, 2017) and implement the graph contrastive loss.

For transductive traffic estimation, we evaluate the performance in terms of the mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2):

$$MAE(\hat{\boldsymbol{X}}, \boldsymbol{X}) = \frac{1}{TN} \sum_{i=0}^{N} \sum_{j=0}^{T} \left|\hat{x}_{i,j} - x_{i,j}\right|, \tag{24}$$

$$RMSE(\hat{\boldsymbol{X}}, \boldsymbol{X}) = \sqrt{\frac{1}{TN} \sum_{i=0}^{N} \sum_{j=0}^{T} \left(\hat{x}_{i,j} - x_{i,j}\right)^2}, \tag{25}$$

$$MAPE(\hat{\boldsymbol{X}}, \boldsymbol{X}) = \frac{1}{TN} \sum_{i=0}^{N} \sum_{j=0}^{T} \frac{\left|\hat{x}_{i,j} - x_{i,j}\right|}{x_{i,j}} \times 100\%, \quad \text{and} \tag{26}$$

$$R^2 = \frac{1}{TN}\sum_{i=1}^{N}\sum_{j=0}^{T}\left(1 - \frac{\left(x_{i,j} - \hat{x}_{i,j}\right)^2}{\left(x_{i,j} - \bar{x}_{i,j}\right)^2}\right). \tag{27}$$

For inductive traffic estimation, the model returns invalid estimations (null, infinity, and negative values) for a few road segments. To address this issue, we suggest new metrics, effective MAE (eMAE), effective MAPE (eMAPE), and effective RMSE (eRMSE), that penalize errors based on the coverage of valid estimation results.

$$eMAE(\hat{\boldsymbol{X}}, \boldsymbol{X}) = \frac{MAE(\hat{\boldsymbol{X}}, \boldsymbol{X})}{C}, \quad eRMSE = \sqrt{\frac{MSE(\hat{\boldsymbol{X}}, \boldsymbol{X})}{C}}, \quad \text{and} \quad eMAPE = \frac{MAPE(\hat{\boldsymbol{X}}, \boldsymbol{X})}{C}, \tag{28}$$

where $C = N_{valid}/NT$ is the proportion of valid estimations, N_{valid}, out of the total number of estimations, and $MSE(\hat{\boldsymbol{X}}, \boldsymbol{X})$ is the square of $RMSE(\hat{\boldsymbol{X}}, \boldsymbol{X})$. We also clip the traffic speed values to the range [0, 200](km/h), and the clipped observations are not included in N_{valid}.

The GTE-GC framework on ten real-world city datasets is implemented with ST-layers $L = 2$, with the size of the hidden dimension $d = 64$. d_1 and d_2 for network features and land-use transformation in the input layer are set to 64 and 32, respectively. The number of heads k is 8 for the multi-head attention in the temporal attention layer. Adam optimizer is used for the training with an initial learning rate of 1e-2 and weight decay of 5e-4. We used $r = 5$ hops to define the neighbors of each node. The parameter to control the contribution of the contrastive loss, α, is set to 1. We train the models for 200 epochs and select the models with the lowest validation set error as the final models. The buffer radii for data preprocessing, β_s and β_f, are set to 100m and 150m, respectively. The models are trained five times, and the mean performance values are reported. For a given city, the training, validation, and testing segments stay the same across all experiments, as in Fig. 4. The models are trained in a PyTorch environment with one NVIDIA RTX A2000 with 12GB memory GPU.

5.3. Results of transductive traffic estimation

Table 3 summarizes the performance of the proposed GTE-GC model and baseline methods across ten cities for the transductive traffic estimation task. In this task, the models are trained and evaluated across different subnetworks within the same city, with the training, validation, and testing sets sharing similar geospatial contexts. The results presented here reflect the performance of ten independently trained models, with one distinct model trained exclusively for each individual study city.

Overall, GTE-GC achieves competitive performance across most cities and metrics. In particular, it yields the best performance (values in bold in Table 3) on at least one metric in London, Munich, Lisbon, and Vienna, highlighting its effectiveness in modeling the spatio-temporal correlations of traffic speed with partial network observations. However, the other variants of GTE also demonstrate competitive performance in the transductive task. GTE, the variant without the graph contrastive loss, consistently yields strong results, achieving near-best performance (values in italic in Table 3) for at least one metric in every city except Madrid, indicating its broad adaptability across diverse urban contexts even without the contrastive objective. GTE-GCN-GC achieves near-best performance in at least one metric in all cities except Zurich and Stockholm. GTE-GCN achieves such results in seven cities, which excludes London, Munich, and Madrid. GTE w/o TE is the only model that does not achieve the best performance for any metric in any city. Still, it remains competitive in three cities: London, Munich, and Madrid. However, the overall performance gap between GTE w/o TE and its temporally informed counterparts (GTE, GTE-GC, and GTE-GCN-GC) confirms the importance of integrating time-of-day information to improve estimation accuracy.

On the other hand, even the relatively simple GCN model performs competitively in the transductive traffic estimation. GCN attains the best performance in Birmingham, London, Munich, Paris, Stockholm, and Madrid, and remains competitive in all cities except Zurich and Lisbon. The competitiveness observed in GTE-GCN, GTE-GCN-GC, and GCN suggests that integrating graph convolution and structured spatial input features can be beneficial in transductive scenarios. These results suggest that even simpler architectures can perform competitively when partial geospatial context is observable during training.

5.4. Results of inductive traffic estimation

In this subsection, we evaluate the performance of the proposed framework under the inductive traffic estimation setting, which requires the model to generalize to entirely unseen networks with different geospatial contexts. Since the optimal source city (network) for generalization is not known a priori, we evaluate all pairwise source-target combinations across the ten cities in Section 5.4.1. Then, comparisons among the baseline models are presented in Section 5.4.2.

5.4.1. Cross-city experiment for source city selection

To discover which source cities serve as effective source domains and provide the best generalization across the other target cities, we conduct comprehensive cross-city experiments by training the model on each city and evaluating it on all other cities. This setting allows us to identify which source cities offer better generalization capability for inductive traffic estimation. We also explore potential drivers of generalization capability.

Table 4 presents the results of this cross-city experiment. Among all source cities, Vienna consistently demonstrates strong generalizability, achieving state-of-the-art performance for most metrics and target cities. Specifically, Vienna achieves either the best or

Table 3
Performance of GTE-GC and baselines for transductive traffic estimation. * **Values in bold** indicate the best performance and *values in italic* with asterisks(*) indicate the performance within a 3% range of the best performance for each metric for each city.

Baseline models	Datasets							
	Zurich				Luzern			
	MAE (km/h)	MAPE (%)	RMSE (km/h)	R^2	MAE (km/h)	MAPE (%)	RMSE (km/h)	R^2
GCN	9.37	23.73	13.15	0.57	*4.53**	*17.59**	*5.95**	*0.44**
GTE w/o TE	9.30	*23.46**	12.10	0.64	4.66	18.07	6.30	0.38
GTE-GCN	**8.81**	**23.01**	**11.58**	**0.67**	**4.44**	**17.29**	**5.92**	**0.45**
GTE-GCN-GC	10.02	24.64	13.78	0.51	*4.46**	*17.49**	*6.02**	0.43
GTE	9.22	*23.39**	11.98	*0.65**	4.59	*17.75**	6.11	0.42
GTE-GC	9.38	24.12	12.25	0.63	4.59	17.87	*6.08**	0.42
	Birmingham				London			
GCN	**7.34**	**31.72**	**9.40**	**0.07**	*6.02**	**21.83**	*8.75**	0.12
GTE w/o TE	7.63	32.98	9.70	0.00	*6.13**	*22.22**	*8.90**	0.09
GTE-GCN	*7.42**	*32.39**	*9.45**	0.05	*6.02**	*21.96**	**8.65**	**0.14**
GTE-GCN-GC	7.59	32.91	*9.62**	0.02	*6.08**	*22.25**	*8.77**	0.11
GTE	7.66	*32.57**	9.77	−0.01	*6.03**	*21.90**	*8.68**	0.13
GTE-GC	7.61	32.82	9.76	−0.01	**6.01**	**21.83**	*8.67**	0.13
	Munich				Paris			
GCN	*6.84**	**20.86**	9.94	0.64	**6.61**	**28.09**	**8.61**	**0.41**
GTE w/o TE	*6.68**	*20.90**	**9.19**	**0.69**	*6.74**	29.08	8.92	0.37
GTE-GCN	*6.68**	*21.04**	*9.28**	**0.69**	*6.76**	29.88	9.01	0.36
GTE-GCN-GC	*6.68**	*21.24**	*9.37**	*0.68**	*6.63**	28.99	*8.76**	0.39
GTE	**6.66**	*21.04**	*9.33**	*0.68**	*6.71**	29.91	8.98	0.36
GTE-GC	*6.67**	*21.09**	*9.29**	**0.69**	*6.78**	30.21	9.02	0.36
	Lisbon				Stockholm			
GCN	11.12	32.06	14.17	0.40	**7.95**	**21.20**	**10.50**	**0.68**
GTE w/o TE	11.52	33.22	15.25	0.31	8.62	22.76	11.61	0.61
GTE-GCN	11.11	32.31	14.31	0.39	8.82	22.18	11.68	0.60
GTE-GCN-GC	*10.92**	*31.15**	14.53	0.37	8.85	22.14	11.77	0.59
GTE	*10.88**	**30.99**	*13.71**	*0.44**	8.56	*21.53**	11.35	0.62
GTE-GC	**10.73**	*31.10**	**13.65**	**0.45**	8.49	*21.57**	11.36	0.62
	Madrid				Vienna			
GCN	**8.14**	**30.86**	**10.68**	**0.42**	7.05	*21.24**	9.31	0.60
GTE w/o TE	*8.29**	*31.42**	*10.83**	*0.41**	6.94	*21.40**	9.13	0.62
GTE-GCN	8.39	32.17	11.19	0.36	*6.78**	*21.13**	*8.93**	**0.64**
GTE-GCN-GC	*8.26**	*31.13**	*10.81**	*0.41**	*6.79**	*21.01**	*8.98**	*0.63**
GTE	8.50	31.93	11.39	0.33	*6.71**	*20.97**	*8.89**	**0.64**
GTE-GC	*8.28**	*31.19**	*10.89**	0.40	**6.66**	**20.93**	**8.84**	**0.64**

near-best eMAE, eMAPE, eRMSE, and R^2 in all cities, except for eMAE in Luzern, eMAPE in Luzern and London, eRMSE in Madrid, and R^2 in Paris, Lisbon, and Madrid. For these cases, the Vienna-trained model achieves the second- or third-best performance except in Madrid. This highlights the Vienna dataset's strong generalization capability across diverse urban contexts as a source dataset. Following Vienna, Zurich also demonstrates high generalization capability. GTE-GC trained in Zurich achieves competitive performance on at least two metrics across six target cities: Luzern, Birmingham, London, Munich, Paris, and Vienna. Finally, Munich and Stockholm also show potential as source cities, yielding competitive performances in three and four target cities, respectively.

Similar trends are also observed in the other GTE variants. GTE, GTE-GCN, GTE-GCN-GC, and GTE w/o TE also demonstrate their best generalization performance when trained on Vienna, Zurich, Munich, or Stockholm, with Vienna being the best option. On the other hand, experiment results for GCN show slightly different results. Although Vienna and Zurich still yield the best performances with the transferred parameters in many cities, training on Munich yields the most consistent estimations. The cross-city experiment results for other models are in Appendix A.

The results of the cross-city experiment highlight the importance of properly selecting source domains for inductive traffic estimation. While the source city selection analysis reveals that Vienna and Zurich offer the most robust inductive performance across target cities, the underlying factors for high generalization capability remain open for further investigation. One hypothesis is the quality of traffic speed data in these cities, which may benefit from higher penetration rates, resulting in more stable and less noisy training labels. This allows the model to better capture stable and robust spatio-temporal relationships rather than overfitting to data irregularities. Although penetration rates for all cities are not available, estimations in our previous study (Luo et al., 2025) showed that Zurich exhibited the highest penetration rate (~11.3%) compared to Madrid (~7.3%) and Luzern (~2.0%), supporting this hypothesis. While London shows a higher average penetration rate of 12.8%, the variance across its network is 2.6 times larger

Table 4
Performance of GTE-GC for inductive traffic estimation across different source-target city combinations. The table reports the performance of the model trained on the cities in each column and tested on the ones in each row. The abbreviations are the first three letters of the cities. * **Values in bold** indicate the best performance and *values in italic* with asterisks(*) indicate the performance within a 3% range of the best performance for each metric for each city.

Test \Train		ZUR	LUZ	BIR	LON	MUN	PAR	LIS	STO	MAD	VIE
ZUR	eMAE		8.64	10.61	13.14	8.23	12.96	11.27	9.14	10.85	**7.60**
	eMAPE		30.62	40.90	39.57	31.16	44.52	44.92	31.96	44.18	**27.67**
	eRMSE		11.61	13.50	16.95	11.20	14.72	14.45	12.40	12.52	**9.97**
	R^2		0.53	0.34	−0.15	0.56	0.13	0.27	0.46	0.40	**0.65**
LUZ	eMAE	**5.37**		10.16	8.94	5.95	8.22	10.37	5.81	8.52	5.58
	eMAPE	**19.52**		36.92	24.33	21.50	26.47	39.04	21.28	29.84	20.30
	eRMSE	**7.40**		13.03	14.11	8.13	10.74	12.58	8.22	10.27	*7.53**
	R^2	**0.69**		0.00	−0.09	0.63	0.37	0.11	0.62	0.58	*0.68**
BIR	eMAE	**8.01**	8.28		10.16	*8.21**	9.96	9.56	8.88	8.53	*8.02**
	eMAPE	*27.50**	28.15		29.19	28.92	29.55	37.58	29.44	29.50	**26.81**
	eRMSE	**10.20**	10.66		13.16	10.85	12.40	12.20	11.60	10.84	**10.20**
	R^2	**0.24**	0.17		−0.27	0.14	−0.13	−0.09	0.01	0.14	**0.24**
LON	eMAE	*5.76**	5.99	7.77		*5.83**	5.96	9.49	*5.73**	7.37	**5.68**
	eMAPE	27.25	28.98	36.69		27.28	**25.67**	45.27	27.09	36.22	26.60
	eRMSE	*7.45**	7.73	10.36		*7.54**	7.81	11.54	*7.53**	8.98	**7.32**
	R^2	0.13	0.07	−0.70		0.11	0.05	−1.10	0.11	−0.19	**0.16**
MUN	eMAE	*8.14**	9.17	9.88	12.97		12.17	10.50	9.97	10.14	**7.91**
	eMAPE	24.72	26.18	32.52	34.66		34.89	35.82	27.63	34.13	**23.51**
	eRMSE	*10.74**	12.64	13.63	15.65		14.05	13.53	14.63	11.85	**10.49**
	R^2	0.58	0.42	0.16	−0.25		0.00	0.34	0.20	0.29	**0.61**
PAR	eMAE	*7.57**	8.19	11.05	8.09	*7.54**		10.51	8.09	9.55	**7.40**
	eMAPE	*34.09**	37.82	48.83	**33.34**	34.54		51.08	36.85	45.73	*33.51**
	eRMSE	*10.07**	10.91	16.44	11.26	**9.95**		13.49	10.91	11.81	*10.16**
	R^2	0.29	0.17	−1.30	0.11	**0.31**		−0.27	0.17	0.13	0.28
LIS	eMAE	11.94	11.99	13.59	15.88	11.92	14.47		**10.91**	12.49	*10.95**
	eMAPE	33.74	33.45	38.94	37.42	33.24	37.00		*30.83**	35.68	**30.14**
	eRMSE	14.82	14.93	18.47	20.92	15.07	17.69		*14.42**	14.83	**14.22**
	R^2	0.32	0.31	−0.16	−0.41	0.28	0.10		0.27	**0.37**	0.30
STO	eMAE	8.51	8.30	11.67	14.36	8.42	11.04	9.24		10.50	**7.77**
	eMAPE	24.25	25.57	35.70	33.36	25.23	30.52	32.11		33.57	**22.59**
	eRMSE	11.00	10.92	15.61	19.68	11.05	14.22	11.82		12.47	**10.52**
	R^2	*0.64**	*0.65**	0.21	−0.26	0.63	0.37	0.56		0.57	**0.66**
MAD	eMAE	9.33	**8.95**	10.88	11.91	9.42	9.95	*9.21**	*8.97**		*9.10**
	eMAPE	35.35	33.66	42.47	34.62	33.67	*32.81**	39.68	34.64		**32.62**
	eRMSE	12.17	*11.72**	15.23	16.94	12.41	13.25	**11.63**	*11.74**		12.00
	R^2	0.37	*0.42**	−0.10	−0.21	0.35	0.26	**0.43**	*0.42**		0.39
VIE	eMAE	**6.83**	7.47	9.43	10.10	*7.02**	8.78	9.09	7.42	8.53	
	eMAPE	**24.40**	26.16	34.34	28.65	25.24	27.47	34.47	26.23	31.04	
	eRMSE	**8.80**	9.89	13.01	13.89	9.46	11.07	11.82	9.82	10.54	
	R^2	**0.54**	0.42	0.02	−0.10	0.47	0.30	0.17	0.43	0.32	

than that of Zurich. Such severe variance may cause instability in the data quality, leading to decreased generalization capability as a source city. In addition, Vienna, Zurich, Munich, and Stockholm exhibit population densities near the median among the ten study cities. Cities with median population densities may exhibit balanced travel behaviors, leading to travel distributions that are close to those of other cities in the latent space, making them more suitable as source domains for generalization across diverse urban forms. Future research will focus on the factors that contribute to effective generalization in the inductive task to develop systematic pre-test criteria or learning-based methodologies for optimal source city selection.

5.4.2. Baseline comparison for the inductive traffic estimation

Table 5 presents a comparison of the proposed GTE-GC model and baselines under the inductive traffic estimation setting, where each model is trained on Vienna and evaluated on all other cities. For estimation in Vienna, the model trained on Zurich is utilized. Consequently, the results reported in this section are derived from two trained models, both of which operate with entirely separate weights from those used in the transductive task.

Across most target cities and metrics, the proposed GTE-GC demonstrates the most accurate results, achieving the lowest error values and the highest R^2 scores. Notably, GTE-GC yields the best performance in at least one metric for eight out of ten cities,

which excludes Paris and Birmingham. In Zurich, London, Munich, Lisbon, and Stockholm, GTE-GC outperforms the other models in all metrics. The proposed model yields the best or near-best performance except eMAPE in Vienna, eRMSE in Paris, and R^2 in Birmingham, Paris, and Madrid.

GTE, GTE-GCN, and GTE-GCN-GC show similar patterns and exhibit relatively robust generalization. GTE yields slightly better estimations than the other two variants. It provides the near-best estimations on at least two metrics in eight cities, while GTE-GCN and GTE-GCN-GC provide such estimations in six cities. This suggests that GTE, which relies on the spatial FFNN without message passing, already captures meaningful structural and contextual information that generalizes well across networks. In addition, incorporating GCN layers may introduce an inductive bias toward the source city's specific topology, which could reduce flexibility when applied to target networks in different geospatial contexts. GTE w/o TE, which lacks temporal encoding, generally performs slightly worse than its temporally-aware counterparts, indicating the advantage of temporal guidance through the encoding technique. GTE-DAT demonstrates substantially degraded performance across the test cases. By relying on domain adversarial training, the model becomes topology-blind, losing the ability to capture local spatial structures. Furthermore, because DAT addresses covariate shifts by aligning marginal feature distributions, it fails to account for label shifts, inadvertently causing negative transfer by overfitting to the source network's specific traffic speed distribution.

Finally, GTE-GC outperforms GTE, highlighting the advantage of the graph contrastive loss in the inductive task. Since the model is trained solely on source networks and evaluated on geographically distinct target networks, direct aggregation of neighbor representations can introduce information from out-of-distribution nodes, potentially degrading the quality of the learned embeddings. The inclusion of graph contrastive loss mitigates this issue by guiding the model to extract representations that are aware of spatial similarities from the node-self's features rather than aggregating all neighbor representations, thereby improving generalization to unobserved networks.

In contrast, GCN demonstrates inconsistent generalization performance. While it performs competitively in Luzern, Birmingham, and Madrid, it struggles significantly in others, exhibiting both high error and negative R^2 values. This suggests that simple GNNs without temporal encoding or structural contrastive supervision are insufficient for generalizing across cities with unobserved geospatial contexts.

5.5. Data ablation

An ablation study is conducted to evaluate the contribution of different input features on the traffic estimation tasks. Using the original *GTE-GC*, which includes structural features (length, FRC, speed limit, number of nearby segments) and 15 land-use categories, as the baseline, we compare five alternative input feature configurations: (i) *land-use*, which only uses land-use features; (ii) *network attributes*, which only uses transportation network attributes; (iii) *+centrality*, which augments the baseline with closeness and betweenness centrality measures; (iv) *+population density*, which supplements the baseline with population density data from Meta,[9] containing globally available, high-resolution estimates of population distribution; and (v) *-nSegment*, which removes the number of nearby segments feature from the baseline.

In the transductive setting (Table 6), the GTE-GC achieves the best performance in five cities: Luzern, London, Munich, Stockholm, and Vienna. The network attributes-only model also performs competitively, yielding lower MAE and/or MAPE than the other baselines in Zurich, Birmingham, Paris, and Lisbon. By contrast, the land-use-only model shows a significant performance degradation. This indicates that land-use features do not provide sufficient context to replace the network attributes, although they may contain complementary information. The introduction of centrality measures and population density does not show a clear advantage over the original input configuration. The model's performance remains at a similar level when population density is adopted, and the centrality measures even degrade the model's performance. On the other hand, the inclusion of the number of nearby segments in the TomTom network dataset enhances the model's performance.

By contrast, in the inductive setting (Table 7), the original input configuration consistently outperforms all others. GTE-GC achieves the best MAE in six cities and the best MAPE in nine cities. The performance degradation in the network attributes model is pronounced in Zurich, Birmingham, Munich, Paris, Stockholm, and Madrid, resulting in eMAE differences exceeding 0.5. The land-use model yields less accurate results as in the transductive task. These results underscore the importance of leveraging both network attributes and land-use features to generalize the model across diverse urban environments. While network attributes alone provide sufficient context to transfer knowledge within the same geospatial context, additional information is necessary to avoid overfitting and transfer knowledge across different geospatial contexts. The contributions of the other data sources are similar to those in the transductive task. The population density-enhanced variant yields competitive performance. However, the model does not offer an advantage considering both the performance and data collection efforts. The centrality measures degrade the model's performance, while the number of nearby segments contributes to the performance.

These findings further emphasize the role of land-use information, particularly in the inductive setting. When paired with network attributes features, land-use features provide sufficient geospatial context to transfer learned spatio-temporal patterns across distinct urban environments. However, when used in isolation, land-use features perform the worst in both transductive and inductive settings, suggesting that the sole use of land-use features is insufficient for reliable traffic estimation without complementary structural data. Meanwhile, the introduction of centrality measures appears to degrade the performance in most cases. This may be due to the model's ability to learn node importance through spatial representations, and the additional centrality measures would be redundant, leading

[9] https://dataforgood.facebook.com/dfg/tools/high-resolution-population-density-maps

Table 5
Performance of GTE-GC and baselines for inductive traffic estimation with *Vienna* as the source network. The estimation of Vienna is done with *Zurich* as the source network. * **Values in bold** indicate the best performance and *values in italic* with asterisks(*) indicate the performance within a 3% range of the best performance for each metric for each city.

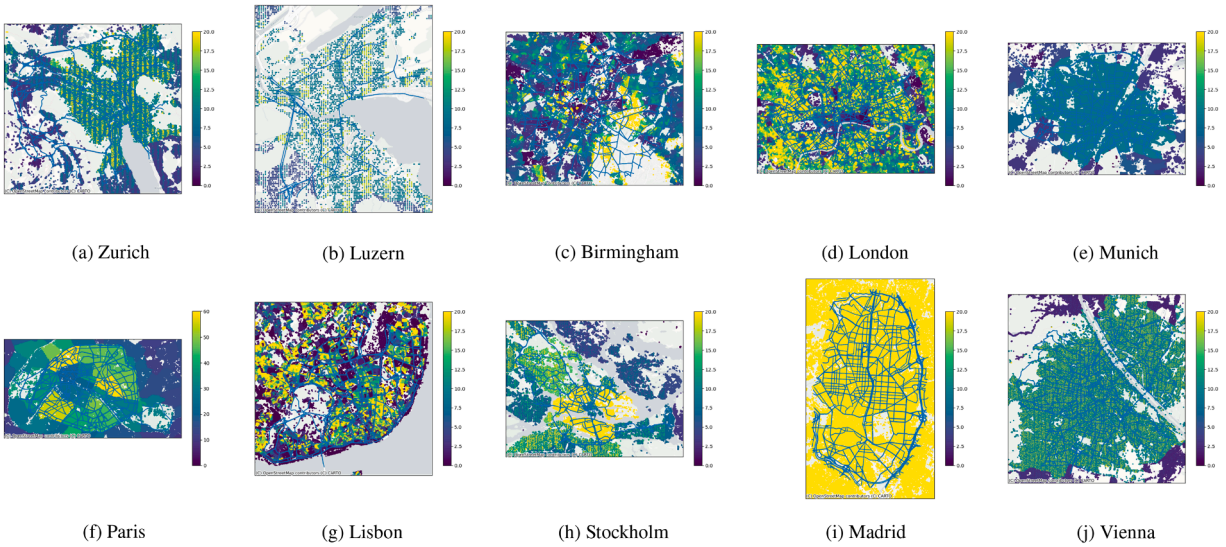
Baseline model	Datasets							
	Zurich				**Luzern**			
	eMAE (km/h)	eMAPE (%)	eRMSE (km/h)	R^2	eMAE (km/h)	eMAPE (%)	eRMSE (km/h)	R^2
GCN	9.61	31.55	19.04	−0.29	*5.73**	**20.00**	7.90	0.65
GTE w/o TE	8.06	28.81	11.15	0.57	5.85	21.29	7.80	*0.66**
GTE-DAT	10.59	33.97	15.61	0.13	7.75	26.28	11.47	0.26
GTE-GCN	7.87	*28.29**	11.09	0.57	**5.57**	*20.17**	*7.56**	**0.68**
GTE-GCN-GC	7.96	*28.23**	11.44	0.54	*5.62**	*20.29**	*7.67**	*0.67**
GTE	*7.75**	*28.07**	*10.24**	0.63	5.78	20.99	7.77	*0.66**
GTE-GC	**7.60**	**27.67**	**9.97**	**0.65**	*5.58**	*20.30**	**7.53**	**0.68**
	Birmingham				London			
GCN	*7.99**	**26.76**	*10.18**	0.24	6.32	28.40	12.93	−1.72
GTE w/o TE	*7.89**	*27.00**	*9.98**	0.27	5.89	27.99	7.92	0.02
GTE-DAT	8.80	30.53	11.83	−0.04	7.79	37.73	10.98	−0.92
GTE-GCN	**7.84**	*26.77**	**9.94**	**0.28**	*5.78**	*27.26**	7.99	0.00
GTE-GCN-GC	*8.02**	*26.98**	*10.21**	0.24	*5.82**	*26.99**	8.23	−0.07
GTE	8.08	*27.02**	10.29	0.22	*5.76**	*26.90**	*7.42**	0.14
GTE-GC	*8.02**	*26.81**	*10.20**	0.24	**5.68**	**26.60**	**7.32**	**0.16**
	Munich				Paris			
GCN	9.42	27.00	18.02	−0.15	7.90	35.20	13.49	−0.44
GTE w/o TE	8.31	24.31	11.46	0.53	*7.39**	34.09	**9.70**	**0.34**
GTE-DAT	10.77	28.81	15.38	0.14	10.06	47.29	13.26	−0.24
GTE-GCN	*8.09**	*23.93**	11.31	0.55	*7.46**	33.94	10.68	0.18
GTE-GCN-GC	8.22	*24.12**	11.53	0.53	*7.42**	*33.27**	10.86	0.13
GTE	*8.09**	*23.84**	10.94	0.58	**7.31**	**32.94**	*9.72**	**0.34**
GTE-GC	**7.91**	**23.51**	**10.49**	**0.61**	*7.40**	*33.51**	10.16	0.28
	Lisbon				Stockholm			
GCN	16.05	42.75	30.82	−2.36	15.84	46.21	34.72	−2.78
GTE w/o TE	*11.27**	31.08	15.38	0.19	8.23	23.68	11.23	0.61
GTE-DAT	13.12	35.06	20.19	−0.49	13.82	35.14	21.46	−0.48
GTE-GCN	11.30	31.23	15.71	0.15	8.15	23.29	11.25	0.61
GTE-GCN-GC	11.43	31.45	16.24	0.09	8.14	*23.22**	11.49	0.59
GTE	*10.96**	*30.18**	*14.23**	**0.30**	*8.00**	*22.90**	*10.74**	0.64
GTE-GC	**10.95**	**30.14**	**14.22**	**0.30**	**7.77**	**22.59**	**10.52**	**0.66**
	Madrid				Vienna			
GCN	9.37	*32.82**	12.42	0.35	8.27	27.00	17.03	−0.71
GTE w/o TE	**9.06**	*33.15**	**11.84**	**0.41**	*7.01**	*24.09**	9.28	0.49
GTE-DAT	10.73	39.20	14.44	0.12	9.32	31.92	13.03	−0.01
GTE-GCN	*9.16**	*33.00**	*12.07**	0.39	*6.98**	24.39	9.42	0.47
GTE-GCN-GC	*9.13**	*32.72**	*12.04**	0.39	7.27	25.33	10.25	0.37
GTE	*9.12**	*32.76**	*11.99**	0.39	*6.86**	**23.63**	**8.78**	**0.54**
GTE-GC	*9.10**	**32.62**	*12.00**	0.39	**6.83**	24.40	*8.80**	**0.54**

Table 6
Results of data ablation study on GTE-GC framework for **transductive** traffic estimation.

Study site	GTE-GC		Land-use		Network attributes		+Centrality		+Population density		-nSegment	
	MAE	MAPE	MAE	MAPE	MAE	MAPE	MAE	MAPE	MAE	MAPE	MAE	MAPE
ZUC	9.38	24.12	14.07	30.51	9.37	**23.61**	9.69	26.43	**9.33**	24.35	9.58	24.85
LUZ	**4.59**	**17.87**	6.58	23.07	4.91	18.33	6.23	23.61	5.51	20.04	5.44	19.36
BIR	7.61	32.82	8.47	37.21	**7.07**	**30.25**	8.42	36.10	7.40	31.80	8.14	35.30
LON	6.01	**21.83**	6.31	23.34	6.25	22.16	**5.98**	22.61	6.06	21.96	6.15	22.77
MUN	**6.67**	**21.09**	9.57	27.35	7.20	21.95	7.11	22.48	6.72	21.39	7.27	22.63
PAR	6.78	30.21	7.69	31.77	6.84	**28.74**	7.00	31.02	**6.75**	30.41	6.91	30.38
LIS	10.73	31.10	13.42	31.58	**10.47**	**29.83**	11.53	31.26	11.03	31.23	11.96	34.08
STO	8.49	**21.57**	14.99	32.48	8.58	21.66	8.38	22.61	**8.22**	21.76	8.49	22.59
MAD	8.28	31.19	10.82	39.52	8.73	31.41	8.45	32.53	**8.15**	**31.07**	8.36	31.71
VIE	6.66	**20.93**	9.87	27.13	7.73	22.16	7.43	24.00	**6.65**	21.03	7.47	23.10

Table 7
Results of data ablation study on GTE-GC framework for **inductive** traffic estimation. The models are trained using the Vienna dataset except for the evaluation on Vienna itself. The city's evaluation is conducted using the models trained using the Zurich dataset.

Study site	GTE-GC		Land-use		Network attributes		+Centrality		+Population density		-nSegment	
	eMAE	eMAPE	eMAE	eMAPE	eMAE	eMAPE	eMAE	eMAPE	eMAE	eMAPE	eMAE	eMAPE
ZUC	**7.60**	**27.67**	10.82	34.14	8.50	28.09	8.34	28.59	7.68	28.76	8.36	31.15
LUZ	**5.58**	20.30	9.86	32.51	5.69	**19.54**	17.11	50.54	5.89	21.92	6.57	24.84
BIR	8.02	**26.81**	8.61	30.25	8.62	27.50	9.46	28.88	8.18	28.51	**7.94**	28.96
LON	**5.68**	**26.60**	7.92	38.83	5.70	26.72	6.95	34.10	6.24	29.47	6.85	33.73
MUN	7.91	**23.51**	10.59	28.71	9.20	25.18	8.44	26.98	**7.86**	23.56	8.70	26.48
PAR	**7.40**	**33.51**	9.98	46.73	8.10	36.55	8.29	39.29	7.62	34.55	7.94	37.93
LIS	10.95	**30.14**	12.26	34.54	11.43	30.78	11.85	32.09	10.88	30.83	**10.87**	31.66
STO	**7.77**	**22.59**	12.60	34.80	8.79	24.35	9.22	25.67	8.22	23.88	8.51	25.59
MAD	9.10	**32.62**	10.52	38.23	9.67	34.91	11.51	34.86	9.48	38.11	**8.97**	34.96
VIE	**6.83**	**24.40**	8.95	28.75	7.15	26.46	7.65	28.03	7.01	25.91	7.35	24.69



(a) Zurich (b) Luzern (c) Birmingham (d) London (e) Munich

(f) Paris (g) Lisbon (h) Stockholm (i) Madrid (j) Vienna

Fig. 6. Visualization of meta high-resolution population density dataset. Colored dots indicate the population density at each 30m × 30m grid, and blue lines indicate the transportation road networks at each city. The population density is almost uniformly distributed in five out of ten cities: Zurich, Luzern, Munich, Madrid, and Vienna. Only Birmingham, Lisbon, and London show heterogeneous population distribution within urban regions.

the model to overfit to the training samples. The inclusion of the number of nearby segments proves particularly effective, reinforcing its utility as a novel spatial descriptor of local road density. This feature, not commonly emphasized in prior studies, offers compact yet meaningful insight into the surrounding network structure and enhances both local and cross-city generalization. Finally, while the inclusion of population density did not yield a consistent advantage in this study, this may be attributed to the uniformity of values in high-density urban areas for the Meta population dataset. In Fig. 6, five out of ten cities show uniform values in central areas, posing a limitation as a distinguishing feature.

5.6. Temporal and structural variations

In this subsection, we analyze the inductive traffic estimation performance of the proposed model across different FRCs and times of day. For comparison, we use i) GTE to investigate the impact of graph contrastive learning on finer spatio-temporal resolution, and ii) GTE w/o TE, which excludes temporal embeddings, to examine its effect on hourly traffic estimation. The evaluation aims to assess how well each model generalizes to unseen cities across road hierarchies and temporal variations. The results are visualized in Fig. 7, where the eMAPEs by FRC (ranging from 0 for major roads to 5 for local roads) and categorized into hourly bins from 5 am to 11 pm are illustrated in heatmap plots. Overall, GTE-GC demonstrates robust performance across all city-FRC combinations except for Zurich-FRC 1, Paris-FRC 0, Paris-FRC 1, and Lisbon-FRC 0 during peak hours. In these cases, however, GTE-GC still outperforms the other variants. These baseline models exhibit unstable performance across all time-of-day bins for several combinations. Specifically, GTE fails to produce accurate estimations for Paris-FRC 0, Paris-FRC 1, and Madrid-FRC 0, while GTE w/o TE fails in Lisbon-FRC 1.

Including GTE-GC, the models exhibit degraded performance for FRC 1, particularly for peak hours in Zurich, Paris, and Lisbon. This may be attributed to the disproportionate FRC distribution of the training dataset. The distribution of FRCs in all cities is skewed

toward local roads, with only 3.77% and 5.44% of segments labeled as FRC 0 and FRC 1, respectively, compared to 19.03%, 21.21%, 25.12%, and 25.43% for FRCs 2 through 5, respectively, on average. However, GTE-GC still achieves higher performance in most FRC-hour bins, suggesting that both temporal encoding and contrastive supervision contribute to model stability in heterogeneous urban scenarios.

5.7. Hyperparameters

Hyperparameter tuning was conducted based on performance in the inductive traffic estimation task. We evaluated sensitivity across five parameters: the number of hops for the graph contrastive loss, $r \in \{2, 4, 5, 6\}$, the number of ST-layers, $L \in \{1, 2, 3\}$, the hidden dimension size, $h \in \{32, 64, 128\}$, the temperature parameter in the contrastive loss, $\tau \in \{1, 2, 3\}$, and the weight of the contrastive loss $\alpha \in \{0.1, 0.5, 1.0, 2.0\}$. The performance of GTE-GC was evaluated with marginal changes from the hyperparameter configuration described in Section 5.2.

Overall, the model demonstrates high robustness to hyperparameter variations. For most configurations, the variation in effective MAE (eMAE) remains within 0.5 km/h of the values obtained using the finalized settings. Notable exceptions include estimations on the Stockholm dataset when the hops are reduced to $r = 2$, the number of layers is changed to $L = 1$ or $L = 3$, or the temperature is increased to $\tau = 3$. Similarly, the eMAPE remains within 1.0% of the baseline in nearly all test cases, with exceptions such as $L = 3$ on London, $L = 1$ on Paris, $\tau = 1$ on Munich, $\tau = 3$ on Stockholm, and $\alpha = 2.0$ on Paris and Lisbon. The largest degradation in eMAE was observed in Stockholm, where setting $\tau = 3$ led to an increase of 0.82 km/h, while setting $\tau = 1$ yielded a 2.81% increase in eMAPE. Ultimately, the finalized hyperparameter configuration reported in Section 5.2 ($r = 5$, $L = 2$, $h = 64$, $\tau = 2$, and $\alpha = 1$) was selected because it demonstrates the most robust and consistent performance across all study areas. Fig. 8 provides a complete summary of this sensitivity analysis.

6. Discussion

6.1. Limitations of the proposed framework

Despite the promising results demonstrated by the proposed GTE-GC framework, we identify three notable limitations that require further investigation.

First, the current framework does not provide a systematic, label-free criterion for selecting source domains before deployment. Although the cross-city experiments identify source cities that empirically yield stronger inductive performance, these results rely on evaluation against target-city labels and therefore cannot be regarded as a reliable source-selection strategy in practical applications where target labels are unavailable. In this sense, source-domain selection remains an empirical sensitivity analysis rather than a rule-based or learning-based component of the proposed framework. Future work should therefore investigate which source-domain properties, target-domain properties, and source-target similarities govern transferability in inductive traffic estimation. A better understanding of these relationships could enable the development of principled source-selection criteria or learning strategies that identify suitable source cities without relying on target labels.

Second, the framework does not yet fully exploit the available multi-city datasets. Although we collected harmonized traffic speed, transportation network, and land-use data across ten European cities, extending the training set to include multiple cities does not consistently improve performance and can even degrade estimation accuracy in both transductive and inductive settings, as shown in Appendix B. This suggests that more sophisticated strategies are needed to integrate multi-city data effectively, particularly by understanding which characteristics of transportation networks and urban contexts contribute to better cross-city generalization. Future work may consider meta-learning approaches to facilitate adaptation across diverse urban contexts, or explore the inclusion of broader geospatial context information, such as country-level attributes or transportation policy indicators, to better guide cross-city transfer. These first two limitations are closely related, as effective expansion of the training set and the development of multi-source learning strategies require a deeper understanding of source-target relationships among different cities.

Lastly, the framework still faces challenges in modeling traffic conditions during traditionally complex periods such as peak hours. Although GTE-GC outperforms the other comparison models in these intervals, estimation accuracy during peak hours remains lower than that of off-peak periods. This may stem from the inherent variability and higher uncertainty associated with congestion dynamics. Addressing this limitation could involve incorporating richer geographic or socioeconomic features that better contextualize travel behavior during high-demand periods. Additionally, extending the model to quantify uncertainty may allow it to handle estimation variability under peak-hour conditions with numerical understanding.

6.2. Practical insights and implications

The proposed GTE-GC framework offers an end-to-end traffic estimation capability, serving as a fast and lightweight complementary tool to traditional, simulation-based traffic modeling approaches. While conventional models remain essential for comprehensive system evaluation and policy simulation, they are often computationally intensive and time-consuming, sometimes requiring several hours or days to complete a single urban scenario analysis. GTE-GC, by contrast, can generate traffic speed estimates in seconds using only static network and land-use inputs. This efficiency makes it particularly valuable in the early stages of planning when multiple design alternatives must be assessed quickly. Additionally, the proposed framework can estimate traffic speed inductively in an end-to-end manner, without requiring sequential procedures and traffic demand inputs, which typically necessitate assumptions,

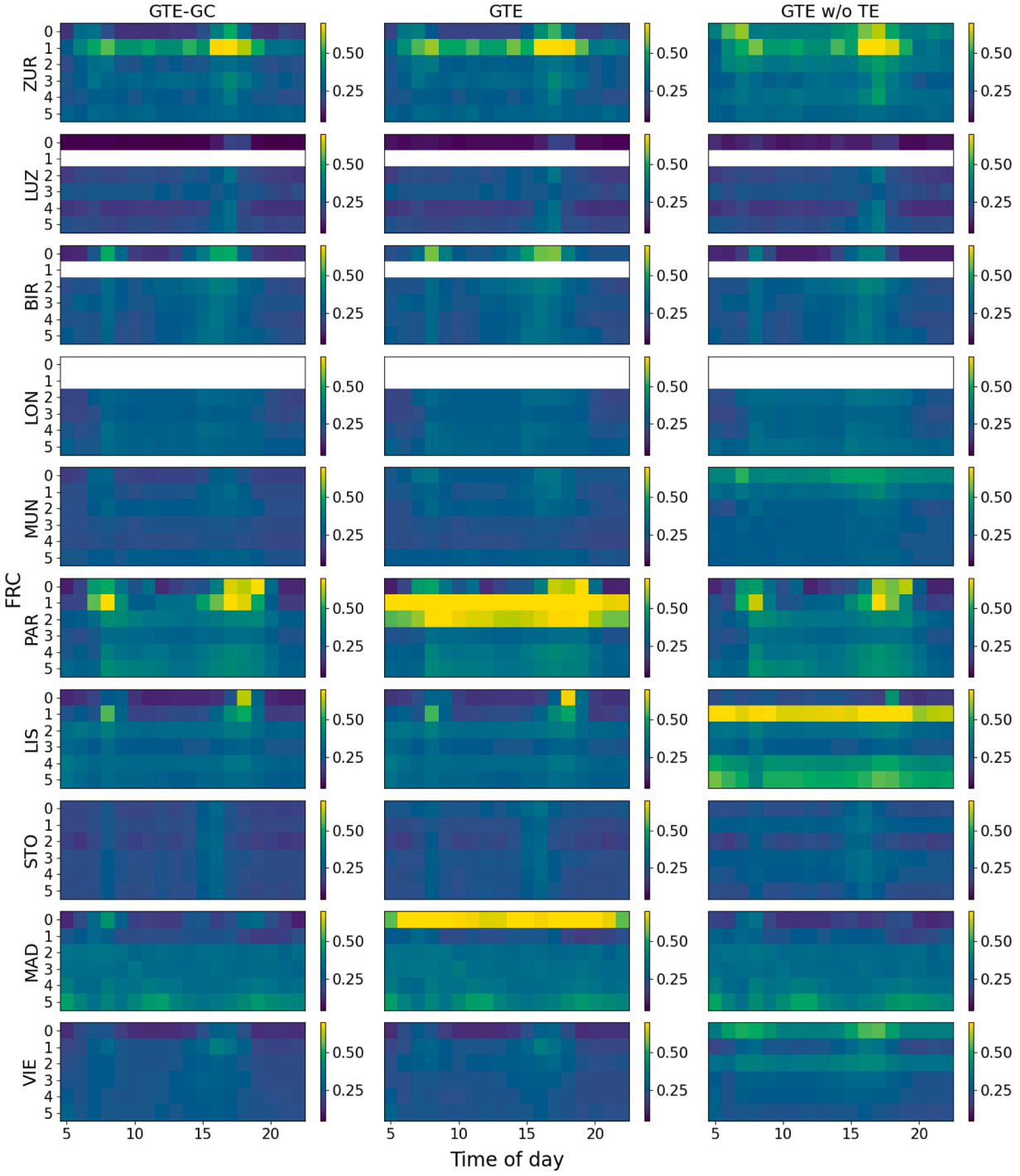
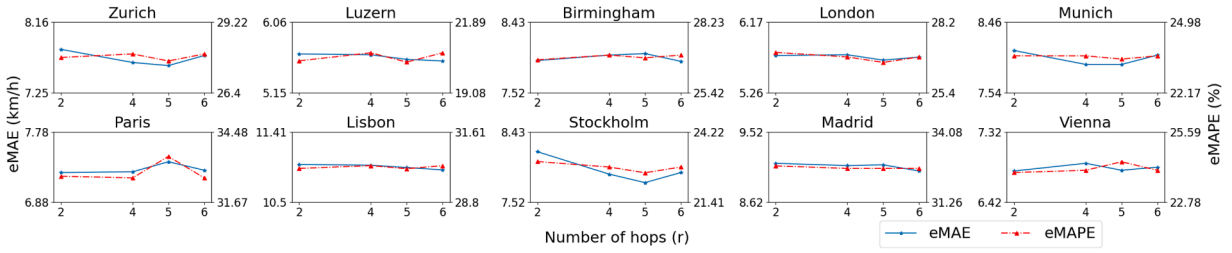
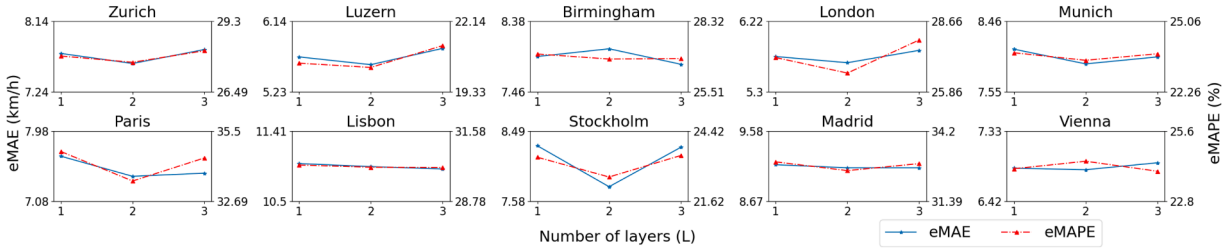


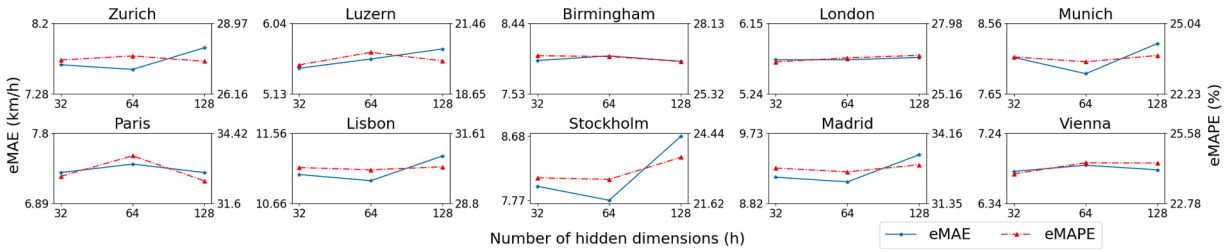
Fig. 7. Visualization of eMAPE for the inductive traffic estimation. All models exhibit degraded performance during peak hours for FRC 0 and 1, which correspond to motorways and international roads. Other than peak hours, however, eMAPE on FRC 0 is much smaller than the error on other FRCs. In addition, GTE-GC tends to consistently yield lower errors during peak hours than its non-temporally encoded counterpart, GTE w/o TE. Note that there are no FRC 1 segments in Birmingham and Luzern and no FRC 0 or 1 segments in London in the original road network datasets.



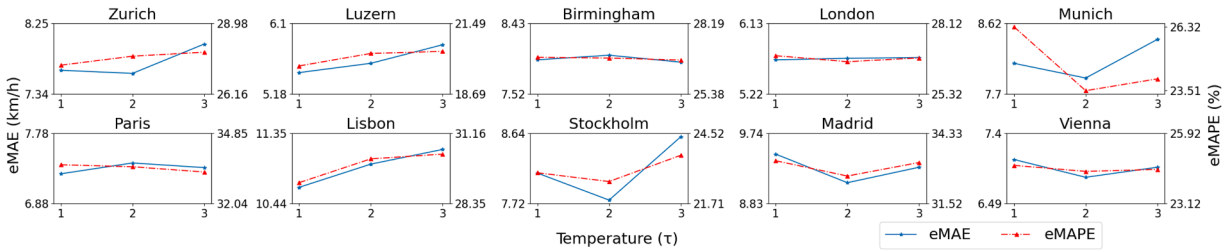
(a) Hyperparameter test for number of hops, $r \in \{2, 4, 5, 6\}$



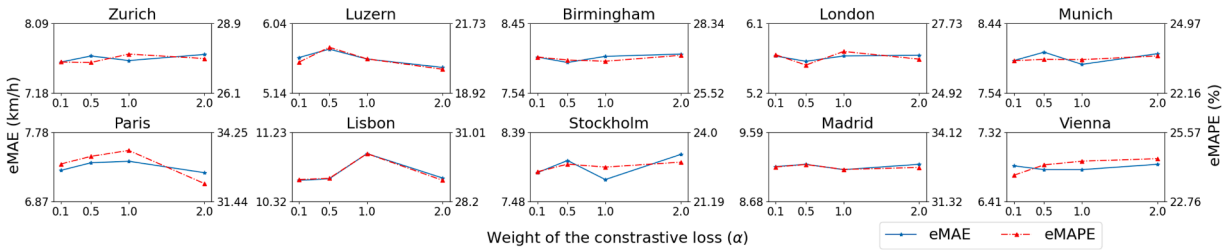
(b) Hyperparameter test for number of layers, $L \in \{1, 2, 3\}$



(c) Hyperparameter test for number of hidden dimensions $h \in \{32, 64, 128\}$



(d) Hyperparameter test for temperature $\tau \in \{1, 2, 3\}$



(e) Hyperparameter test for the weight of the contrastive loss $\alpha \in \{0.1, 0.5, 1.0, 2.0\}$

Fig. 8. Results of the hyperparameter sensitivity analysis evaluated via eMAE and eMAPE. No single alternate configuration universally outperforms the selected baseline. The final hyperparameters are set to $r = 5$, $L = 2$, $h = 64$, $\tau = 2$, and $\alpha = 1.0$, as defined in Section 5.2.

surveys, or synthetic modeling during the planning process. By relying solely on structural and contextual data, GTE-GC enables practitioners to obtain sound traffic speed estimates without prior traffic observations or demand models. This enables rapid screening of candidate road network layouts, allowing practitioners to identify promising configurations or potential issues before committing to resource-intensive high-fidelity models. This framework is applicable to planning new urban developments as well as expanding or modifying existing transportation networks, where historical traffic data may be unavailable or unreliable.

The findings also caution against the assumption that more data necessarily improves model performance. As shown in the data ablation analysis in Section 5.5, adding population density or centrality measures did not yield performance gains and, in some cases, even degraded accuracy. This suggests that when such data are coarse, lack variability, or do not sufficiently reflect the real world, they may introduce noise or redundancy that hampers learning. For example, population data with uniform distributions in dense urban cores may provide limited utility for differentiating traffic conditions at the segment level. Practitioners should therefore weigh the representational fidelity and relevance of additional inputs before integrating them into the modeling process. Overloading the model with insufficiently informative data may not only slow down computation but also obscure the core spatio-temporal patterns the model is designed to learn.

7. Conclusion and future work

In this study, we proposed GTE-GC, a generalizable end-to-end deep learning framework to address two traffic estimation problems: which involves estimation of traffic statistics on unobserved segments within the same network used for training, leveraging shared geospatial context; and inductive traffic estimation, which requires estimation in entirely different networks, where the model has not seen any part of the target network during training. The framework estimates traffic speed in road networks using only transportation topology and land-use context, without relying on historical traffic or demand data. The model introduces a novel spatio-temporal encoding architecture that combines a spatial FFNN, temporal self-attention, and graph contrastive learning to support both transductive and inductive traffic estimation tasks. We conducted extensive experiments using traffic speed, transportation road network, and land use datasets across ten European cities, showing that GTE-GC yields competitive performance in transductive settings and superior generalization in inductive tasks, demonstrating the model's ability to learn transferable representations from static transportation and urban features. In addition, the dataset ablation study confirmed the contribution of each data source, while also showing that certain features, such as centrality measures and population density, offered limited advantage. Finally, the model demonstrated robust performance across various road types and time periods, outperforming the other baselines. As a result, the proposed framework offers a practical and lightweight tool to complement traditional simulation-based approaches for early-stage transportation planning.

In future work, we aim to expand the scope of this study to include more diverse data sources and non-recurring traffic patterns. While this work focuses on traffic speed, traffic flow remains a fundamental measure for understanding transportation systems. Although we only analyze traffic speed due to data availability, GTE-GC is capable of estimating both speed and flow, as in previous ST-GNN studies (Li et al., 2018; Shin and Yoon, 2024). A promising avenue for future research involves extending this framework to incorporate traffic flow, which would enable a more comprehensive assessment of urban mobility. Specifically, we plan to explore the integration of multi-source datasets by combining the broad spatial coverage of probe-vehicle speed data with the rich, point-wise flow measurements from stationary sensors. Developing a reliable coupling framework to fuse these heterogeneous sensing sources will be a key step toward achieving a more robust representation of network-wide traffic conditions. Finally, while the current framework is deliberately scoped to estimate typical, recurring weekday traffic profiles, a critical next frontier is the disruption-aware estimation of traffic dynamics under non-recurring events such as incidents, severe weather, or special events, and the assessment of network vulnerability to such disruptions. The primary bottleneck in modeling these anomalies is the sheer difficulty of obtaining comprehensive supervision data across all possible spatial and temporal disruption scenarios. To overcome this data scarcity, future research could explore advanced generative techniques (Choi et al., 2025; Nie et al., 2025) for realistic traffic disruption data augmentation and synthesis, as well as for few-shot estimation of traffic states. By utilizing these generative approaches, future extensions of our work could simulate and evaluate network-wide responses to complex, non-recurring disruptions without relying on exhaustive historical datasets.

CRediT authorship contribution statement

Yuyol Shin: Writing - original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Wei Ma:** Writing - review & editing, Validation, Supervision, Methodology, Conceptualization; **Monica Menendez:** Writing - review & editing, Visualization, Validation, Supervision, Project administration, Funding acquisition, Formal analysis, Data curation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT and Grammarly to refine the tone and check grammar. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Acknowledgments

Yuyol Shin and Monica Menendez are partially supported by the NYUAD Center for Interacting Urban Networks (CITIES), funded by Tamkeen under the NYUAD Research Institute Award CG001. Wei Ma is supported by the Research Grants Council of the Hong Kong Special Administrative Region, China (Project No. PolyU/15227424). We thank Dr. Daniel Bramich and Dr. Jesper Skottfelt for defining the temporal and spatial coverage of the TomTom dataset and managing its retrieval.

Data availability

The data used in the study are partially public and partially confidential. Links to available data are presented in the paper.

Appendix A. Evaluation of networks' generalization capabilities

The performance of the cross-city experiment for evaluating the generalization capability of different source networks is presented in Tables A.1–A.5 for GTE, GTE-GCN, GTE-GCN-GC, GTE w/o TE, and GCN. For all GTE variants, the Vienna network demonstrated the best capability as the source network, followed by Zurich and Munich. Since GTE-GAT strictly requires concurrent access to both the source and target networks during the training phase to align their respective feature distributions, conducting an exhaustive cross-city evaluation for GTE-DAT would necessitate independently training 90 distinct models (10 source cities × 9 target cities). To maintain tractability while still providing a rigorous comparison, we designated Vienna as the single source network and trained separate GTE-DAT models for each of the remaining nine target cities.

For GCN, Zurich and Munich yielded the most cases of top-level performance. However, the model trained on Vienna showed the most robust performance, consistently producing eMAPES less than 50%. The other source networks showed unreliable performance in several regions, such as Lisbon and Stockholm.

Table A.1
Performance of **GTE** for inductive traffic estimation across different source-target city combinations.

Test \Train		ZUR	LUZ	BIR	LON	MUN	PAR	LIS	STO	MAD	VIE
ZUR	eMAE		8.71	11.76	13.04	*7.97**	12.54	10.98	9.2	10.95	**7.75**
	eMAPE		30.85	44.25	39.46	30.55	43.10	43.28	31.17	44.59	**28.07**
LUZ	eMAE	**5.58**		16.09	8.94	*5.69**	8.44	9.63	5.77	8.44	5.78
	eMAPE	**19.73**		57.14	24.42	20.92	27.17	36.31	21.03	30.16	20.99
BIR	eMAE	8.26	8.36		10.08	**7.98**	9.69	9.25	8.75	8.39	*8.08**
	eMAPE	*27.31**	28.47		29.04	28.57	29.21	36.35	28.88	29.46	**27.02**
LON	eMAE	*5.71**	*5.90**	7.98		5.86	5.93	8.58	**5.68**	7.60	*5.76**
	eMAPE	*26.35**	28.36	37.16		27.93	**25.80**	41.32	26.66	36.53	26.90
MUN	eMAE	*8.30**	9.33	9.96	12.93		11.82	10.22	9.89	10.00	**8.09**
	eMAPE	*24.51**	26.60	32.47	34.58		33.89	34.44	26.93	34.10	**23.84**
PAR	eMAE	*7.36**	8.15	16.25	8.07	7.60		9.89	8.00	9.73	**7.31**
	eMAPE	**32.46**	37.31	69.20	33.49	35.43		48.24	36.18	45.18	*32.94**
LIS	eMAE	12.20	12.05	13.26	15.69	12.07	14.42		*11.08**	12.87	**10.96**
	eMAPE	33.14	33.61	37.80	37.17	33.52	36.55		*30.68**	36.79	**30.18**
STO	eMAE	8.80	8.44	15.35	14.15	8.40	10.75	9.31		10.22	**8.00**
	eMAPE	23.97	25.62	46.05	33.03	25.31	29.83	31.74		32.75	**22.90**
MAD	eMAE	9.37	*9.02**	10.70	11.85	*9.20**	9.86	*9.20**	**8.97**		*9.12**
	eMAPE	33.72	33.86	41.15	34.75	33.50	**32.21**	39.15	33.66		*32.76**
VIE	eMAE	**6.86**	7.58	10.98	9.98	*6.87**	8.83	8.88	7.41	8.59	
	eMAPE	**23.63**	26.34	37.82	28.50	25.01	27.17	33.62	25.84	31.55	

Table A.2
Performance of **GTE-GCN** for inductive traffic estimation across different source-target city combinations.

Test \Train		ZUR	LUZ	BIR	LON	MUN	PAR	LIS	STO	MAD	VIE
ZUR	eMAE		8.76	12.22	15.05	*8.01**	15.72	12.8	9.6	13.38	**7.87**
	eMAPE		30.98	45.72	45.96	30.30	53.06	49.66	32.17	53.28	**28.29**
LUZ	eMAE	**5.40**		12.31	10.69	5.77	10.46	12.18	5.97	11.21	5.57
	eMAPE	**19.61**		43.22	29.46	20.83	33.29	44.70	21.54	38.34	*20.17**
BIR	eMAE	8.10	8.33		10.27	*8.08**	9.95	10.14	8.72	8.51	**7.84**
	eMAPE	*27.33**	28.31		29.47	28.6	29.74	39.6	29.03	30.52	**26.77**
LON	eMAE	**5.78**	6.02	7.85		*5.84**	*6.04**	10.19	*5.85**	8.13	**5.78**
	eMAPE	27.15	28.99	37.38		27.51	**26.10**	48.15	27.81	39.23	27.26
MUN	eMAE	8.37	9.30	10.10	14.30		14.29	12	10.35	11.45	**8.09**
	eMAPE	25.06	26.54	33.16	38.92		41.38	40.40	27.93	38.85	**23.93**
PAR	eMAE	7.86	8.25	11.02	8.26	**7.46**		11.99	8.18	10.14	**7.46**
	eMAPE	*34.61**	38.10	49.23	**33.94**	*34.15**		57.21	37.41	47.39	**33.94**
LIS	eMAE	13.34	12.68	13.40	17.73	12.72	17.60		**11.16**	14.02	*11.30**
	eMAPE	37.12	34.99	39.91	42.04	35.42	46.5		**30.72**	40.56	*31.23**
STO	eMAE	8.78	8.50	12.91	15.03	8.58	12.36	9.98		12.71	**8.15**
	eMAPE	24.67	25.78	38.87	35.33	25.36	33.9	34.26		40.25	**23.29**
MAD	eMAE	9.54	**8.96**	10.03	12.09	9.33	10.08	9.44	*9.01**		*9.16**
	eMAPE	35.13	*33.15**	40.69	34.93	*33.05**	**32.89**	40.17	34.01		*33.00**
VIE	eMAE	*6.98**	7.63	10.09	11.07	**6.97**	9.74	10.53	7.46	9.40	
	eMAPE	**24.39**	26.51	36.48	31.24	*24.93**	30.21	38.85	26.14	34.21	

Table A.3
Performance of **GTE-GCN-GC** for inductive traffic estimation across different source-target city combinations.

Test \Train		ZUR	LUZ	BIR	LON	MUN	PAR	LIS	STO	MAD	VIE
ZUR	eMAE		8.87	13.06	15.51	*8.17**	14.75	11.89	9.60	14.10	**7.96**
	eMAPE		31.68	48.46	47.35	31.29	49.68	46.51	31.74	55.12	**28.23**
LUZ	eMAE	*5.68**		12.24	10.74	5.85	9.82	10.45	5.92	12.76	**5.62**
	eMAPE	*20.79**		43.89	30.22	21.46	31.18	39.26	21.01	43.75	**20.29**
BIR	eMAE	*8.10**	8.34		10.65	*8.06**	9.66	9.59	8.89	8.51	**8.02**
	eMAPE	*27.63**	28.45		30.76	28.98	29.08	37.8	29.21	29.90	**26.98**
LON	eMAE	6.10	6.07	7.92		6.04	5.92	9.63	**5.69**	7.87	*5.82**
	eMAPE	28.79	29.33	37.72		28.83	**25.53**	46.15	26.79	38.00	26.99
MUN	eMAE	*8.42**	9.28	10.46	14.78		13.89	11.25	10.5	12.07	**8.22**
	eMAPE	25.20	26.65	34.23	40.24		40.17	37.5	27.87	40.51	**24.12**
PAR	eMAE	8.17	8.28	10.93	8.13	8.00		10.90	7.98	10.12	**7.42**
	eMAPE	36.40	38.30	50.49	*33.52**	37.38		53.36	35.91	47.36	**33.27**
LIS	eMAE	13.48	13.16	14.07	17.32	12.04	17.23		**11.35**	14.93	*11.43**
	eMAPE	38.17	37.61	42.15	41.72	33.76	45.85		**30.83**	43.11	*31.45**
STO	eMAE	9.05	8.45	12.6	15.75	8.64	12.00	9.33		13.40	**8.14**
	eMAPE	25.79	26.14	39.17	37.62	25.99	32.7	32.64		41.84	**23.22**
MAD	eMAE	9.40	**8.94**	10.20	11.82	9.25	9.91	*9.13**	*9.00**		*9.13**
	eMAPE	35.05	*33.52**	41.45	34.64	*33.45**	**32.65**	39.32	33.64		*32.72**
VIE	eMAE	*7.27**	7.61	9.85	10.94	**7.07**	9.51	9.30	7.43	9.84	
	eMAPE	**25.33**	26.63	35.98	31.34	*25.57**	29.58	35.21	*25.83**	35.21	

Table A.4
Performance of **GTE w/o TE** for inductive traffic estimation across different source-target city combinations.

Test \Train		ZUR	LUZ	BIR	LON	MUN	PAR	LIS	STO	MAD	VIE
ZUR	eMAE		8.90	13.79	17.03	*8.08**	15.65	12.79	9.54	14.85	**8.06**
	eMAPE		31.4	50.31	51.31	30.73	51.41	49.14	34	56.72	**28.81**
LUZ	eMAE	**5.55**		14.38	11.77	*5.59**	10.61	11.58	6.24	14.35	5.85
	eMAPE	**19.70**		51.72	33.83	20.42	33.82	42.67	22.94	50.69	21.29
BIR	eMAE	8.27	8.42		10.69	*8.10**	10.37	10.04	8.98	8.75	**7.89**
	eMAPE	*27.37**	28.65		30.49	28.53	30.30	38.81	30.91	30.64	**27.00**
LON	eMAE	**5.78**	6.03	7.85		*5.79**	6.03	9.56	6.21	8.22	*5.89**
	eMAPE	27.09	29.12	37.33		27.42	*25.41**	45.67	29.35	39.23	27.99
MUN	eMAE	*8.41**	9.3	10.78	16.28		14.60	11.94	9.22	12.58	**8.31**
	eMAPE	*24.72**	26.73	34.92	44.54		41.18	39.71	27.82	41.27	**24.31**
PAR	eMAE	*7.49**	8.37	12.35	8.49	7.65		11.33	8.56	10.37	**7.39**
	eMAPE	**33.44**	38.54	55.72	35.01	35.29		54.14	39.07	48.40	*34.09**
LIS	eMAE	13.03	13.04	14.43	18.20	12.31	17.38		**10.75**	15.08	11.27
	eMAPE	35.09	36.55	43.38	44.08	33.99	45.07		**30.34**	43.45	*31.08**
STO	eMAE	9.12	8.83	14.08	15.85	*8.46**	12.34	9.50		14.61	**8.23**
	eMAPE	24.73	26.71	43.83	37.75	25.27	33.61	32.64		45.58	**23.68**
MAD	eMAE	9.47	9.04	10.31	11.92	9.19	10.39	9.31	**8.70**		9.06
	eMAPE	*33.97**	*33.69**	41.92	34.66	**33.09**	*33.60**	39.30	*33.23**		*33.15**
VIE	eMAE	*7.01**	7.75	10.45	11.75	**6.95**	10.15	10.43	7.90	10.08	
	eMAPE	**24.09**	26.88	37.23	33.45	25.03	30.79	38.30	27.67	35.47	

Table A.5
Performance of **GCN** for inductive traffic estimation across different source-target city combinations.

Test \Train		ZUR	LUZ	BIR	LON	MUN	PAR	LIS	STO	MAD	VIE
ZUR	eMAE		10.01	30.18	34.77	**7.98**	54.59	13.73	10.67	55.27	9.61
	eMAPE		32.60	90.33	95.95	**29.97**	146.8	49.77	36.26	159.56	31.55
LUZ	eMAE	**5.45**		24.21	19.34	*5.52**	24.87	10.8	6.70	56.72	5.73
	eMAPE	**19.09**		84.07	57.74	19.83	77.95	39.39	23.62	196.36	20.00
BIR	eMAE	*8.19**	*8.18**		10.58	*8.02**	10.12	9.57	8.56	10.07	**7.99**
	eMAPE	*27.26**	*27.18**		30.08	27.98	30.15	37.11	29.44	34.87	**26.76**
LON	eMAE	6.39	7.13	9.19		**6.01**	7.09	10.42	6.82	19.73	6.32
	eMAPE	*28.31**	31.9	41.24		**27.62**	29.17	48.00	31.55	82.48	*28.40**
MUN	eMAE	**8.93**	10.25	20.18	30.95		48.42	12.79	10.18	41.83	9.42
	eMAPE	**25.50**	28.33	55.56	78.92		123.21	41.37	30.35	111.77	27.00
PAR	eMAE	**7.26**	7.65	17.14	8.25	*7.34**		10.09	8.34	25.5	7.90
	eMAPE	**30.52**	34.55	72.75	33.3	33.33		48.08	38.64	104.54	35.2
LIS	eMAE	31.88	32.94	32.16	25.42	18.27	47.62		**10.70**	46.10	16.05
	eMAPE	97.19	98.81	97.15	64.43	54.51	139.87		**30.53**	138.79	42.75
STO	eMAE	44.72	45.38	127.20	98.07	28.13	147.93	**9.14**		55.71	15.84
	eMAPE	139.30	142.79	391.33	286.23	86.25	451.16	**30.65**		158.21	46.21
MAD	eMAE	9.42	9.30	11.95	11.90	9.26	9.82	9.04	**8.67**		9.37
	eMAPE	33.75	33.77	46.38	34.64	*32.60**	**31.62**	37.75	33.02		32.82
VIE	eMAE	8.27	8.42	15.76	15.85	**7.23**	20.66	10.82	8.38	25.98	
	eMAPE	27.00	27.51	50.84	44.29	**25.37**	59.00	38.41	28.51	76.29	

Appendix B. Traffic estimation tasks with multiple source cities

Table B.1 shows the performance of transductive and inductive traffic estimation tasks using multiple cities in the training stage.

Table B.1
Results with expanded datasets.

	Transductive estimation							
Train \Test	Zurich				Vienna			
	MAE	MAPE	RMSE	R^2	MAE	MAPE	RMSE	R^2
Transductive	9.38	24.12	12.25	0.63	6.66	20.93	8.84	0.64
Zurich & Vienna	15.86	30.21	22.24	−0.21	10.27	26.84	14.59	0.03
Zurich, Munich, & Vienna	14.64	28.65	20.43	−0.02	9.69	26.06	13.57	0.16
	Inductive estimation							
Train \Test	Luzern				Madrid			
	eMAE	eMAPE	eRMSE	R^2	eMAE	eMAPE	eRMSE	R^2
Vienna	5.58	20.30	7.53	0.68	6.83	24.40	8.80	0.54
Zurich & Vienna	8.30	26.64	12.34	0.15	10.81	38.89	14.71	0.09
Zurich, Munich, & Vienna	8.01	26.29	11.71	0.24	10.58	38.81	14.17	0.15

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